

Beyond Attack Success Rate: Temporal Logit Observability for LLM Safety Failures

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Abstract

Attack Success Rate (ASR) evaluates each jailbreak with a single yes/no label at the end of generation, telling us *whether* a failure happened but not *how* it unfolded. Two attacks that produce equally harmful outputs may have followed completely different paths, and ASR cannot tell them apart. We make those hidden paths observable from logits alone. *Temporal Logit Observability* (TLO) is a training-free diagnostic that watches a compliance–refusal margin during decoding and places each model–attack condition on a calibrated 2D plane. By design, this plane is most informative exactly where ASR is least informative: among attacks that succeed for genuinely different reasons. Across four aligned LLMs and three jailbreak paradigms, attacks with nearly identical ASR land at clearly different points on the plane: the same model can fail through different temporal patterns. The geometry matches refusal-direction probes from hidden states on most conditions, with one model showing the limit of our fixed-lexicon approach. A simple early-stop rule derived from TLO cuts successful jailbreaks by more than half, without false alarms on plain benign queries. Safety evaluation should report *when* and *how* a failure unfolds, not only *whether* it occurred. TLO makes the first two observable from logits alone.

1 Introduction

Attack Success Rate (ASR) evaluates aligned large language models: a binary judgment of whether the final response refused or complied with a harmful prompt [20, 19]. ASR is operationally appealing and has become the de facto standard for jailbreak evaluation [8, 21, 2, 3]. But ASR is a single number computed at the *end* of generation: it treats decoding as a black box, recording the final label but not the decoding trajectory.

This gap is a structural property of the metric: ASR by construction observes only the final response, discarding the trajectory by which it was produced. As shown in Figure 1(a), distinct attacks against the same harmful question can produce nearly indistinguishable final responses that ASR labels identically. The same ASR-success label can hide three fundamentally different processes: refusal evidence may have appeared briefly and been suppressed; refusal may have emerged but too late to matter; or refusal may never have been triggered at all [11]. Because ASR collapses these into one number, model–attack pairs with identical ASR may require different defenses that current evaluation cannot distinguish.

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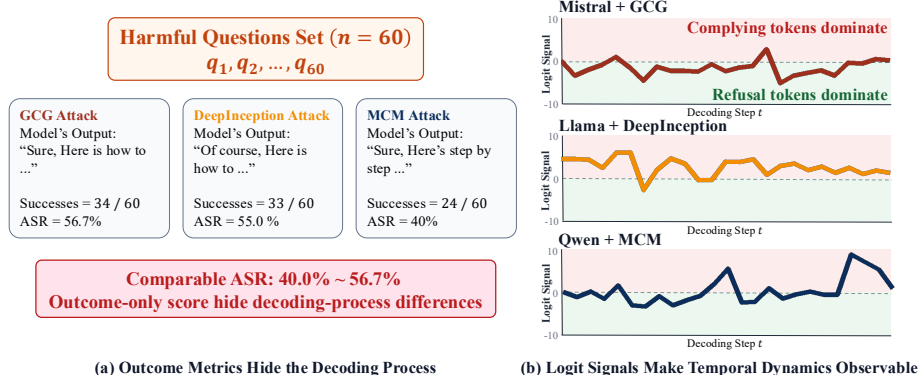


Figure 1: **ASR collapses temporally distinct safety failures.** (a) Distinct attacks against the same harmful question can yield equally compliant final responses, indistinguishable under ASR. (b) The model–attack conditions follow temporally distinct logit trajectories, recoverable as TLO’s per-step compliance–refusal margin.

Existing approaches that recover this hidden structure rely on hidden-state probes [1, 29, 27], which require activation access and model-specific instrumentation. Output logits are far more widely available, yet existing logit-level safety work has focused on intervention, such as steering or decoding modification [17], rather than diagnosis. This leaves an open question: *can the temporal structure of safety failures hidden by ASR be recovered from logits alone, without hidden-state access?*

Our approach. We introduce *Temporal Logit Observability* (TLO), a training-free, logit-only diagnostic protocol that converts decoding-time trajectories into a calibrated two-dimensional coordinate. TLO measures a compliance–refusal margin (the Logit-Margin Score, LMS) at each decoding step, summarizes it along pre-generation and generation-time axes, and calibrates each summary against harmful and attack-formatted benign references via Relative Position (RP). The resulting RP-plane coordinate $c = (RP_A, RP_B)$ is comparable across models, attacks, and prompt formats, and exposes *when* safety evidence concentrates without inspecting any internal state. TLO complements ASR rather than replacing it. By construction, RP-plane displacement compresses with $(1 - \text{ASR})$: RP carries the primary signal in mid-ASR conditions, where ASR alone is least informative about *why* attacks succeed, while ASR carries the load when failures are rare.

What TLO recovers. Across four aligned LLMs (Llama, Mistral, Qwen, Gemma) and three jailbreak paradigms (MCM, GCG, DI), TLO recovers structure that ASR systematically discards (Figure 1(b)). First, attacks with nearly identical ASR occupy clearly distinct locations in the RP plane: within attack DI, Llama and Qwen separate at $D_{RP} = 0.572$ despite an ASR difference of only 3.3%, and across attack family the separation reaches $D_{RP} = 1.014$ at a 1.7% ASR difference; the same attack family (GCG) further ranges from the RP origin on Qwen to a large generation-time displacement on Llama (§4.2). Second, the underlying logit trajectory separates successful from failed jailbreaks within the first few generated tokens, and aligns with hidden-state refusal directions on 10 of 12 model–attack conditions, with model-specific scope clearly delimited (§4.3). Third, the geometry supports decoding-time intervention: a simple t_{cross} -derived early-stop rule reduces aggregate ASR from 39.6% to 13.1% with 0.0% false-positive rate on format-free benign queries (§4.4). These observations are stable across decoding randomness, lexicon perturbations, judge variation, and stricter activation-timing rules (§4.5).

Contributions.

1. **Calibrated temporal coordinates from logits alone.** The RP-plane coordinate $c = (RP_A, RP_B)$ summarizes the LMS trajectory along pre-generation and generation-time axes, calibrated against harmful and attack-formatted benign references. The $(1 - \text{ASR})$ bound on RP displacement is a deliberate complementarity with ASR (§3).
2. **Model–attack interactions invisible under ASR.** Across 12 conditions, the same attack family produces qualitatively different temporal signatures across models, and the same model handles

different attacks via different temporal mechanisms (§4.2). The recovered geometry is supported by hidden-state refusal directions on 10 of 12 conditions, with model-specific scope made explicit (§4.3).

3. **Decoding-time intervention.** A simple t_{cross} -derived halt rule reduces aggregate ASR from 39.6% to 13.1% with 0.0% false alarms on format-free benign queries, demonstrating that the calibrated geometry drives effective decoding-time monitoring (§4.4).

2 Preliminaries

We review ASR as the outcome-level metric that motivates TLO, then introduce the logit-only decoding notation used throughout the paper.

Evaluation at the final response. Attack Success Rate (ASR) formalizes jailbreak evaluation as a final-response outcome metric [20, 30, 19]. By aggregating final-response labels, ASR records whether an attack succeeded but discards the token-level trajectory by which the model moved toward refusal or compliance.

Decoding setup and access. We operate in a logit-only access setting. Consider an autoregressive language model [24, 4] generating response tokens $y_1, \dots, y_T$ given context $\mathbf{x} = (x_1, \dots, x_n)$. We compute one prompt-only diagnostic logit vector $\ell_0 \in \mathbb{R}^{|\mathcal{V}|}$ at the final prompt position before any response token is generated. During generation, ℓ_t for $t = 1, \dots, T$ denotes the score vector returned immediately before sampling response token y_t . The decoding index thus has one pre-generation position and T generation-step positions; we write the observed score trajectory as $S_{0:T}$ after defining LMS below. TLO uses only token-level logits to observe when safety-related evidence appears during decoding, without gradients, hidden states, architectural changes, or decoding-time intervention. It requires full logits or targeted lexicon-token log-probabilities; standard top- k outputs are insufficient (Appendix F.10). Hidden-state analyses serve as activation-based baselines for validation (§4.3).

Logit-Margin Score. We use the Logit-Margin Score (LMS) to measure the compliance–refusal balance at each decoding step. Li and Liu [17] introduce this margin as a *steering* signal; we use the same observable for passive temporal diagnosis, and the calibration and analysis built on top of it form our contribution. We aggregate compliance and refusal lexicons $\mathcal{L}_{\text{cmp}}, \mathcal{L}_{\text{ref}}$ (Appendix B.1) into mean scores $\mu_{\text{cmp}}(t)$ and $\mu_{\text{ref}}(t)$, and define LMS as:

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (1)$$

Positive S_t indicates compliance-dominant logits; negative S_t indicates refusal-dominant logits. We use lexicon-internal top- k aggregation with $k = 10$; sensitivity is reported in Appendix F.6.

3 Temporal Logit Observability

Given an LMS trajectory $S_{0:T}$, the *Temporal Logit Observability* (TLO) protocol converts decoding-time logit observations into calibrated 2D coordinates without hidden-state access or model modification. TLO and ASR are complementary by design: TLO is most informative when attacks often succeed (where ASR alone says little about *why*), while ASR is most informative when attacks rarely succeed. TLO proceeds in three stages: (i) a per-step temporal logit signal (§3.1); (ii) activation-timing summaries such as crossing time and sign reversal (§3.2); and (iii) calibration against harmful and attack-formatted benign references via Relative Position (§3.3). Figure 2 summarizes the pipeline.

3.1 Temporal Logit Signal

TLO computes the Logit-Margin Score S_t from Eq. 1 at each decoding step, yielding a trajectory $S_{0:T}$. From this trajectory, we record three compliance–refusal summaries that drive activation timing and RP calibration.

S_0 is the *pre-generation observation*, computed at the final prompt position before any response token is generated. It captures whether the input context has already shifted the model’s logit state toward compliance before decoding begins.

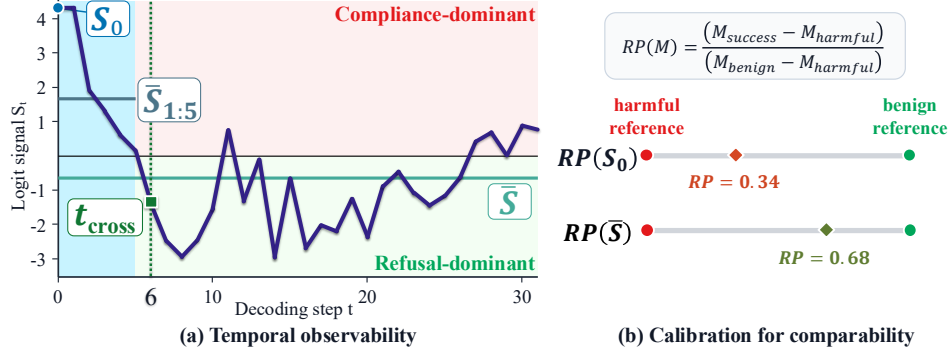


Figure 2: **Operationalizing TLO.** (a) A real harmful-sample logit trajectory illustrates the external time series of compliance–refusal balance during generation. (b) Relative-position calibration expresses each temporal observation relative to harmful and attack-formatted benign references, making conditions comparable without hidden-state access.

$\bar{S} = \frac{1}{T} \sum_{t=1}^T S_t$ is the *generation-time observation*, computed as the mean logit signal over the T generation steps. A prefix variant $\bar{S}_{1:w} = \frac{1}{w} \sum_{t=1}^w S_t$ provides an early-window summary. S_0 and $\bar{S}$ jointly form the axes of the RP-calibrated plane in §3.3.

3.2 Activation Timing Signals

Activation-timing signals measure when the LMS trajectory first shows refusal-dominant evidence.

$t_{\text{cross}} = \min\{t \geq 1 : S_t < 0\}$ records the first generation step at which refusal dominates ($t \geq 1$ excludes the pre-generation position S_0). If no crossing occurs within the generated sequence, t_{cross} is right-censored. For ROC analysis, censored samples are tied at the maximum crossing rank within the analysis horizon, not ordered by response length. A stricter n -consecutive-step variant is verified in Appendix F.2.

Sign reversal ($S_0 > 0 \wedge \bar{S} < 0$) records a sustained shift from an initially compliance-dominant trajectory to refusal dominance during generation. We compute sign reversal only on harmful prompts, as benign sign patterns are format-dependent rather than safety-driven (Appendix B.3).

3.3 Relative-Position Calibration

Raw logit observations cannot be directly compared across models, attacks, or prompt formats: a positive S_t on Qwen and on Llama do not mean the same thing. We therefore calibrate each observation through **Relative Position (RP)**, which expresses successful attacks relative to harmful and attack-formatted benign distributions:

$$\text{RP}(M) = \frac{M_{\text{success}} - M_{\text{harmful}}}{M_{\text{benign}} - M_{\text{harmful}}}, \quad (2)$$

where M_{success} , M_{harmful} , and M_{benign} are condition means over successful attacks, all harmful samples, and attack-formatted benign references. $\text{RP} = 0$ indicates harmful-like behavior; $\text{RP} = 1$ indicates benign-like behavior; values outside $[0, 1]$ are reported as observed inversions rather than clipped. The all-harmful and attack-formatted benign anchors avoid any need for failed-jailbreak samples while controlling for the attack wrapper. Anchor sensitivity, denominator audits, and a rank-based variant are reported in Appendix C.2.

Applying Eq. 2 to S_0 yields RP_A ; applying it to $\bar{S}$ yields RP_B . We treat the resulting two-dimensional coordinate $c = (\text{RP}_A, \text{RP}_B)$ as the primary calibrated observation, and define RP-plane divergence between conditions a and b as the Euclidean distance:

$$D_{\text{RP}}(a, b) = \|c_a - c_b\|_2. \quad (3)$$

D_{RP} is measured in RP-normalized units: each axis is scaled by the harmful-to-benign reference gap of its temporal summary, so one unit equals the harmful–benign anchor distance for that summary. Higher D_{RP} indicates greater calibrated temporal separation, even when final ASR is similar.

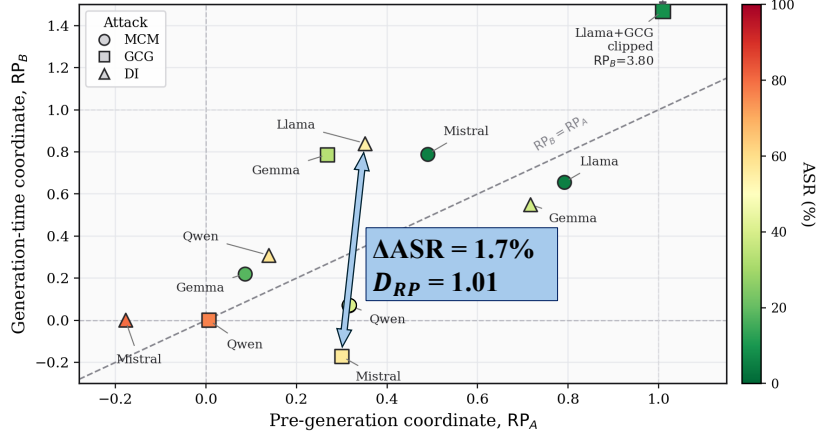


Figure 3: **Conditions with similar ASR can occupy distinct RP-plane locations.** Each point is a model–attack condition with coordinate $c = (RP_A, RP_B)$ that captures pre-generation and generation-time calibrated logit position; marker shape denotes attack family and color denotes ASR under the primary Llama-Guard judge. The diagonal marks $RP_B = RP_A$ for visual reference. Llama+GCG has $RP_B = 3.80$ and is clipped for readability; all analyses use unclipped coordinates.

What RP measures, by design. Because the harmful anchor pools all harmful samples, $M_{\text{success}} - M_{\text{harmful}} = (1 - \text{ASR})(M_{\text{success}} - M_{\text{failed}})$, so RP-plane displacement is bounded by $(1 - \text{ASR})$ on each axis. This formalizes the complementarity introduced at the start of §3: as ASR rises, RP shrinks toward zero by design. High-ASR conditions therefore carry their primary signal in ASR itself, while mid-ASR conditions are where RP separates attacks that ASR cannot.

Reading the RP plane. The continuous coordinate c is the primary statistic. For readability, we attach significance-filtered axis annotations: a label-permutation test on the displacement $r = \|c\|_2$ with Benjamini–Hochberg FDR at $q = 0.05$ identifies conditions with reliably non-zero displacement (Appendix C.3). For displaced conditions, we mark the dominant axis as **Pre-generation Bias** (PGB) when $|RP_A| > |RP_B|$, and **In-generation Bias** (IGB) when $|RP_B| > |RP_A|$. Conditions failing the permutation test are annotated as **Null-generation Bias** (NGB), meaning no reliable RP-axis displacement under the calibrated protocol. These labels mark axis dominance throughout the paper; axis-margin and small-success caveats are reported in Appendix C.4.

4 Experiments

The experiments answer three questions about TLO. **(Q1)** Does the calibrated RP plane recover the model–attack structure that ASR collapses (§4.2)? **(Q2)** Are the underlying logit signals informative and aligned with hidden-state safety representations (§4.3)? **(Q3)** Can the recovered geometry drive action during decoding (§4.4)? We then summarize robustness checks (§4.5).

4.1 Setup

We evaluate TLO on a 4×3 grid of four open instruct models (Llama [13], Mistral [16], Qwen [26], Gemma [12]) and three jailbreak paradigms, with 60 harmful JailbreakBench prompts [6] per condition under greedy decoding. The attack grid covers multi-turn context manipulation (MCM; Appendix B.2), suffix optimization (GCG [30]), and template-based semantic jailbreaks (DeepInception, DI [18]). For RP calibration, each condition also includes 60 attack-formatted benign references that preserve the wrapper but replace the harmful request, yielding 720 harmful generations and 720 attack-formatted benign generations. Attack success is judged by Llama-Guard-3-8B [15]; auxiliary judges and a human audit are reported in Appendix A–G.

Table 1: **RP-plane coordinates separate model–attack conditions that ASR collapses.** Each row reports a condition’s calibrated 2D position, displacement magnitude r , permutation significance, and the early-stop probe’s intervention response (§4.4). PGB = Pre-generation Bias; IGB = In-generation Bias; NGB = Null-generation Bias (high-ASR range where displacement shrinks by design).

Model	Attack	RP-plane coordinates and annotation						Behavioral probe ($w = 5$)		
		ASR	RP_A	RP_B	r	p_{perm}	Annotation	ASR_{orig}	ASR_{probe}	ΔASR
Llama	GCG	8.3%	1.01	3.80	3.94	<.001	IGB	8.3%	1.7%	6.7
	MCM	5.0%	0.79	0.66	1.03	<.001	PGB	5.0%	5.0%	0.0
	DI	55.0%	0.35	0.84	0.91	<.001	IGB	55.0%	8.3%	46.7
Mistral	GCG	56.7%	0.30	−0.17	0.35	<.001	PGB	56.7%	53.3%	3.3
	MCM	5.0%	0.49	0.79	0.93	.056	<i>NGB</i>	5.0%	5.0%	0.0
	DI	80.0%	−0.18	0.00	0.18	.506	<i>NGB</i>	80.0%	0.0%	80.0
Qwen	MCM	40.0%	0.32	0.07	0.32	.271	<i>NGB</i>	40.0%	40.0%	0.0
	GCG	76.7%	0.01	0.00	0.01	.987	<i>NGB</i>	76.7%	8.3%	68.3
	DI	58.3%	0.14	0.31	0.34	.074	<i>NGB</i>	58.3%	5.0%	53.3
Gemma	GCG	33.3%	0.27	0.79	0.83	.003	IGB	33.3%	11.7%	21.7
	MCM	18.3%	0.09	0.22	0.24	<.001	IGB	18.3%	18.3%	0.0
	DI	38.3%	0.72	0.55	0.90	<.001	PGB	38.3%	0.0%	38.3
		Aggregate	39.6%	13.1%						26.5

4.2 Q1: RP-Plane Geometry Recovers ASR-Hidden Model–Attack Interactions

Our central claim is that conditions with similar ASR can occupy distinct locations in the RP plane. We use the continuous coordinate $c = (RP_A, RP_B)$ and pairwise D_{RP} as primary observations; PGB/IGB/NGB labels (Eq. 7) are descriptive axis annotations. Figure 3 visualizes the 12 condition coordinates with ASR as color, and Table 1 reports coordinates, displacement magnitude, permutation p -values, RP-plane annotations, and the early-stop probe response (the probe itself is reported in §4.4).

ASR-neighboring conditions occupy distinct RP-plane locations. Among nine model–attack pairs with ASR within 5%, the median calibrated separation is $D_{RP} = 0.572$ with stratified-bootstrap 95% CI [0.489, 0.924] (Appendix C.2, Table 7). The cleanest within-attack pair, Llama+DI vs. Qwen+DI (ASRs 55.0% vs. 58.3%, $\Delta = 3.3\%$), separates at $D_{RP} = 0.572$ [0.237, 1.026], showing that the same attack on different models produces distinct calibrated geometries even at near-equivalent ASR. The strongest ASR-neighbor pair, Llama+DI vs. Mistral+GCG (ASRs 55.0% vs. 56.7%, $\Delta = 1.7\%$), reaches $D_{RP} = 1.014$ [0.726, 1.351], adding attack identity to model identity as a second source of mechanism difference. For reference, within-condition stratified-bootstrap intervals on the displacement magnitude have a median 95% CI width of about 0.6 RP units (Appendix C.3, Table 11); both pair separations exceed this typical within-condition spread. ASR-neighbor conditions therefore have nearly identical safety outcomes but qualitatively different decoding-time geometries: structure that ASR collapses by design.

Attack family does not determine RP-plane location. The same attack family occupies different RP-plane locations across models. GCG ranges from Qwen near the RP origin (0.01, 0.00) (expected from the $(1 - \text{ASR})$ bound at ASR 76.7%) to Llama’s large generation-time displacement (1.01, 3.80) at low ASR 8.3%, with Mistral and Gemma occupying intermediate regions (PGB and IGB respectively). Llama+GCG’s $RP_B = 3.80$ exceeds the $[0, 1]$ reference interval because successful samples lie beyond the benign anchor on the generation-time axis; this inversion is stable under rank-based RP and bootstrap resampling (Appendix C.2, C.3). Defenses tuned to a single attack family but uniformly applied across models will therefore see condition-dependent effects that ASR cannot predict.

Model identity does not determine RP-plane location. The same model occupies different RP-plane locations under different attacks. Llama places MCM (0.79, 0.66) (PGB) apart from DI (0.35, 0.84) (IGB), and Gemma places DI (0.72, 0.55) (PGB) apart from MCM (0.09, 0.22) (IGB). A single per-model defense without attack-aware adaptation therefore mismatches the underlying temporal structure.

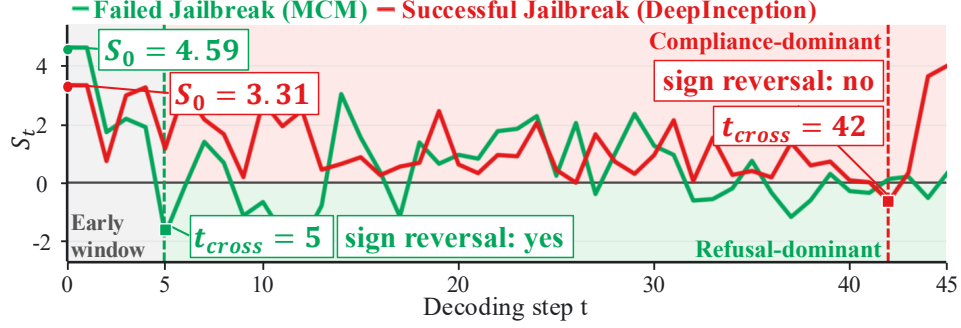


Figure 4: **TLO marks safety-activation timing on the trajectory.** For the same harmful question on Mistral under two attack wrappers, the failed MCM jailbreak crosses early ($t_{\text{cross}} = 5$) and shows sign reversal, while the successful DI jailbreak delays crossing ($t_{\text{cross}} = 42$) without sign reversal.

Reliable conditions span the grid. Of the 12 conditions, 7 show statistically reliable RP-plane displacement under FDR-corrected permutation, and these 7 span all 4 models and all 3 attacks. The remaining 5 are either high-ASR (NGB by design; e.g., Mistral+DI at ASR 80%, Qwen+GCG at ASR 76.7%) or have low success counts that produce wide bootstrap intervals (Llama+MCM, Mistral+MCM, $n_{\text{succ}} = 3$). An independent entropy-based check confirms NGB reflects the high-ASR range rather than lexicon insensitivity (Appendix C.6).

4.3 Q2: The Logit Signal Is Informative and Tracks Hidden-State Safety Representations

The RP geometry of §4.2 rests on the LMS trajectory and on the assumption that this trajectory carries safety-relevant information. We verify both: the trajectory predicts safety outcomes within the first few generated tokens, and aligns with hidden-state refusal directions on most conditions, with model-specific scope made explicit.

The trajectory separates outcomes within the first five tokens. Failed jailbreaks typically cross into refusal dominance early, while successful jailbreaks delay or avoid that transition (Figure 4). Within the first five generated tokens, successful and failed jailbreaks separate at Cohen’s $d > 1.0$ ($p < 10^{-38}$, $n = 720$): the safety outcome is observable in the logit margin long before the response is complete. $\bar{S}_{1:5}$ achieves ROC-AUC [10] = 0.771 [0.737, 0.804] and t_{cross} achieves 0.784 [0.750, 0.816]. A trained 13-feature logistic regression on the same trajectory features reaches AUC = 0.826 (Appendix F.3), so the fixed-form summaries used here serve as an interpretable lower bound on extractable trajectory information. Failed jailbreaks exhibit sign reversal $5.2\times$ more often than successful ones (40.0% vs. 7.7%), and successful jailbreaks remain compliance-dominant much more often (92.3% vs. 49.7%); delayed or absent refusal activation is therefore the logit-level signature TLO observes (token-rank evidence in Appendix D.1).

The trajectory aligns with hidden-state refusal directions, with model-specific scope. We fix the hidden layer at $\sim 50\%$ network depth (L50), compute a per-model difference-of-means refusal direction from paired harmful/benign references without success labels, and project decoding-time hidden states. The expected S_{t-z_t} correlation ρ is negative, summarized over steps 0–4 (full protocol, layer sweep, and probe-AUC comparisons in Appendix E).

Alignment strength varies by model. On Llama, all three attacks reach $|\rho| \geq 0.70$, and the modal cross-correlation lag is -1 : the logit margin moves *one decoding step earlier* than the refusal-direction projection, consistent with autoregressive feedback in which the $t-1$ output token modulates $\mathbf{h}_t$ through the input context. Mistral and Gemma preserve the expected sign with moderate strength (peaking at $|\rho| = 0.46$ for Mistral+GCG and $|\rho| = 0.50$ for Gemma+DI). Qwen marks a model-specific boundary: it reverses the expected sign under MCM and DI, while only Qwen+GCG retains it. The Fisher- z mean across Qwen conditions is approximately zero, a known limit of the fixed-lexicon protocol on Qwen rather than a contradiction of the cross-model pattern. Model-specific or probe-derived lexicons recover more evidence in such cases (Appendix F.8, F.9).

Across the 12 conditions, 10 show the expected sign between S_t and refusal-direction projections (Table 2); layer-sweep checks at L25, L50, and L75 confirm this is not a layer-tuned result (Appendix E.2).

Table 2: **Hidden-state alignment summary.** Expected-sign alignment between S_t and fixed-L50 hidden-state projections across model–attack conditions.

	Llama			Qwen			Mistral			Gemma		
	MCM	GCG	DI	MCM	GCG	DI	MCM	GCG	DI	MCM	GCG	DI
Hidden-state alignment												
Strong alignment	✓	✓	✓									
Aligned					✓		✓	✓	✓	✓	✓	✓
Boundary				✓		✓						

4.4 Q3: From Observation to Action via RP-Aware Early Stopping

The RP-plane geometry can drive action during decoding. We derive a halt rule directly from t_{cross} : halt when $t_{\text{cross}} > 5$ and replace the continuation with a standard refusal, using the fixed early-activation window of §4.3.

Aggregate effect. With $w = 5$, aggregate ASR drops from 39.6% to 13.1% ($\Delta = 26.5\%$), and the response varies systematically by RP-plane region (PGB $\Delta = 13.9\%$, IGB $\Delta = 18.8\%$, NGB $\Delta = 40.3\%$; Appendix G.3). The largest drops occur in NGB conditions, consistent with the complementarity of §3: where RP geometry is uninformative by design, ASR carries the primary signal and a uniform halt rule is most effective.

Benign behavior. Format-free benign queries had 0.0% false positives across all 12 conditions. Attack-formatted benign references, where benign prompts are deliberately wrapped in the attack template, share the same temporal cues used to detect attacks; an S_0 -gated variant that requires a pre-generation harmful signal reduces false halts on these stress prompts from 46.0% to 12.6% (Appendix G.4).

4.5 Robustness

The temporal and RP-plane structure persists across decoding randomness ($\text{ICC} = 0.90\text{--}0.94$), stricter confirmed-crossing rules, a trained 13-feature classifier ($\Delta\text{AUC} = 0.04$ over fixed t_{cross}), non-lexical controls (entropy, logit norm), lexicon perturbations ($|\Delta\text{AUC}| \leq 0.018$ across five variants), metric decomposition (~ 0.16 AUC margin gain over single-component variants), and data-driven lexicon induction (Appendix F.1–F.8). TLO is therefore not a surface refusal-token detector or a compliance-onset heuristic.

5 Related Work

TLO connects three lines of work: final-response safety evaluation, hidden-state safety representations, and inference-time defenses. These respectively measure jailbreak outcomes, expose internal safety signals, and intervene during decoding, but none provides a calibrated temporal diagnostic from standard logits. TLO fills this gap as a passive logit-level observation method that tracks when, and whether, safety evidence becomes detectable during generation.

Safety evaluation. Safety training through human preference learning has produced models that resist harmful requests, motivating evaluation frameworks that measure this resistance at scale [28, 23]. Benchmarks such as HarmBench [20] and AdvBench [30], together with red-teaming efforts [11, 22], standardize jailbreak evaluation largely through ASR. ASR is practical and widely adopted, but it reduces the decoding process to a final-response label. TLO complements these outcome-level metrics by tracking the compliance–refusal balance during decoding and calibrating it into RP-plane coordinates, exposing the cross-condition geometry that ASR cannot represent.

Safety representations. Representation-level work shows that safety behavior leaves measurable traces in hidden states. Refusal-direction analysis identifies linear directions associated with refusal, representation engineering uses such directions for behavioral control, and response-safety representations rank outputs by safety quality [1, 29, 9]. Yin et al. [27] shows that hidden-state refusal scores can change at token-level time scales, but this evidence requires activation access. Li and Liu [17] use a compliance–refusal logit gap for steering; we adopt the same observable for passive temporal diagnosis, and contribute the RP calibration that makes the resulting trajectory comparable across models, attacks, and prompt formats. Activation-based detectors are stronger-access baselines when hidden-state access is available; TLO targets the lower-access full-logit setting that frontier-style auditing increasingly demands.

Inference-time detection and defense. Inference-time detection and defense methods use generation-side signals to identify or reduce jailbreak risk without changing model weights. Early-logit detectors show that first-pass or first-token output distributions can efficiently flag jailbreak inputs [5, 7]; these methods produce per-prompt verdicts and target input classification. SafeDecoding [25] re-weights token probabilities toward safe continuations, while CARE [14] uses a detect–rollback loop to halt harmful generations. TLO targets a different task. Rather than a per-prompt verdict or a decoding policy, it is a calibrated cross-condition diagnostic: it tracks how compliance–refusal evidence evolves across the full decoding window, and exposes model–attack interactions that per-prompt detectors and uniform decoding policies do not see.

6 Discussion and Limitations

TLO offers a calibrated logit-only diagnostic for safety failures. We state several limits explicitly to scope how it should be interpreted.

Access and coverage. TLO requires full logits or targeted logit-bias probing—standard top- k outputs at $k = 20$ are insufficient (Appendix F.10)—and uses a fixed English compliance–refusal lexicon supported by probe-derived rank checks, perturbations, and data-driven induction (Appendix F.9, F.5, F.8). It is therefore most directly applicable to open-weight English auditing; per-language or per-model lexicons can be plugged into the same pipeline, while API-restricted deployment via lightweight distilled monitors is left to future work, and adaptive manipulation of the logit channel itself is outside our threat model.

Format effects and observational scope. Temporal observability responds to attack formatting as well as harmful intent: format-free benign queries are stable, but attack-formatted benign references can trigger the same temporal cues used to detect attacks, motivating S_0 -gated variants for deployment-aware monitors (Appendix G.4). More broadly, TLO is observational—it makes safety-failure dynamics visible from logits but does not identify underlying circuits or causal mechanisms; this limit is intentional, since observation is what the field currently lacks for ASR-collapsed evaluation. Future work should connect TLO observations to specific intervention mechanisms, expand to multilingual lexicons and frontier model scales, and develop standardized audit protocols that preserve the observation–causation distinction.

7 Conclusion

ASR records whether an attack succeeded but not how. *Temporal Logit Observability* (TLO) makes the missing dimension observable from logits alone: a training-free 2D RP-plane coordinate, designed to be complementary to ASR rather than to replace it. Across four aligned LLMs and three jailbreak paradigms, the RP plane separates ASR-neighboring conditions that ASR collapses, agrees with hidden-state refusal directions on 10 of 12 conditions with model-specific scope made explicit, and drives a uniform early-stop rule that cuts aggregate ASR by 26.5% without false alarms on plain benign queries. This first step, from observation to actionable monitoring without attack-specific tuning, opens the more direct question of whether attack-aware, RP-aware defenses can do better than the uniform halt rule reported here. Safety evaluation should report *when* and *how* a failure unfolds, not only *whether* it occurred.

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A Evaluation Grid and Signal Anchors

The evaluation grid supports the main observability claim with 12 model–attack conditions. We evaluate Llama-3.1-8B-Instruct [13], Mistral-7B-Instruct-v0.3 [16], Qwen2.5-7B-Instruct [26], and Gemma-2-9B-Instruct [12] against multi-turn context manipulation (MCM), GCG [30], and Deep-Inception (DI) [18]. Each condition contains 60 harmful JailbreakBench prompts [6] and 60 attack-formatted benign references. Llama-Guard-3-8B [15] provides the primary success labels; Appendix G records the auxiliary-judge refreshes and their condition-level disagreements.

Table 3: Evaluation grid and calibrated signal anchors. Harmful columns report ASR and condition-mean signal values; benign columns report attack-formatted reference anchors for RP calibration.

Model	Attack	Harmful			Benign	
		ASR	S_0	$\bar{S}$	S_0	$\bar{S}$
Llama	MCM	5.0%	1.45	−1.42	6.38	+1.25
	GCG	8.3%	0.62	−0.99	4.29	−0.53
	DI	55.0%	4.43	−0.11	6.50	+0.72
Mistral	MCM	5.0%	4.51	−0.31	5.25	+0.67
	GCG	56.7%	3.14	+0.16	4.76	−0.52
	DI	80.0%	3.04	+0.43	3.10	+0.69
Qwen	MCM	40.0%	11.97	−0.12	12.89	+1.43
	GCG	76.7%	7.13	+0.71	12.80	−0.60
	DI	58.3%	7.86	+1.18	12.52	+1.63
Gemma	MCM	18.3%	4.67	+0.47	9.74	+2.32
	GCG	33.3%	4.46	+0.68	8.63	+1.07
	DI	38.3%	4.63	+0.89	6.69	+1.49

A.1 Authoritative 12-Condition Result Ledger

All main-text aggregate results use the latest 12-condition Llama-Guard grid ($n = 720$ harmful samples). This appendix reports the same 12-condition values whenever an aggregate or per-condition number is used as evidence for the main claims.

Table 4: Primary 12-condition success-prediction summary ($n = 720$ harmful samples; Llama-Guard success labels).

Metric	ROC-AUC	95% CI
t_{cross}	0.784	[0.750, 0.816]
$\bar{S}_{1:5}$	0.771	[0.737, 0.804]
$\bar{S}$	0.758	[0.724, 0.793]
Keyword refusal baseline	0.748	[0.715, 0.780]
$\bar{S}_{1:3}$	0.723	[0.687, 0.759]
S_0	0.680	[0.640, 0.718]

The pooled AUC in Table 4 is a summary over heterogeneous model–attack conditions, not a claim that every condition is equally separable. Table 5 makes this heterogeneity explicit: Qwen+GCG is near chance for all reported logit summaries, while several Llama conditions are near ceiling.

A.2 ASR-Neighboring Motivation

ASR-neighboring model–attack pairs can be temporally non-equivalent. Figure 5 retains raw trajectory examples as qualitative motivation. The main text reports the calibrated RP-plane version of this comparison in Figure 3 and Appendix C.2.

Table 5: Per-condition AUC heterogeneity in the same 12-condition grid. Values are oriented so larger AUC indicates stronger success/failure separation for that condition.

Model	Attack	n_{succ}	ASR	S_0	$\bar{S}_{1:3}$	$\bar{S}_{1:5}$	$\bar{S}$	t_{cross}	Keyword
Gemma	DI	23	38.3%	0.792	0.760	0.719	0.806	0.792	0.832
Gemma	GCG	20	33.3%	0.723	0.672	0.693	0.725	0.629	0.725
Gemma	MCM	11	18.3%	0.700	0.688	0.696	0.857	0.528	0.510
Llama	DI	33	55.0%	0.699	0.717	0.750	0.934	0.799	0.825
Llama	GCG	5	8.3%	0.935	0.971	0.964	0.920	0.987	0.855
Llama	MCM	3	5.0%	0.924	0.918	0.918	0.977	0.614	0.649
Mistral	DI	48	80.0%	0.542	0.535	0.634	0.521	0.667	0.625
Mistral	GCG	34	56.7%	0.745	0.791	0.827	0.631	0.705	0.693
Mistral	MCM	3	5.0%	0.737	0.725	0.708	0.942	0.570	0.947
Qwen	DI	35	58.3%	0.619	0.627	0.627	0.574	0.653	0.723
Qwen	GCG	46	76.7%	0.511	0.517	0.550	0.516	0.519	0.540
Qwen	MCM	24	40.0%	0.536	0.532	0.520	0.595	0.685	0.500
Macro average				0.705	0.705	0.717	0.750	0.679	0.702

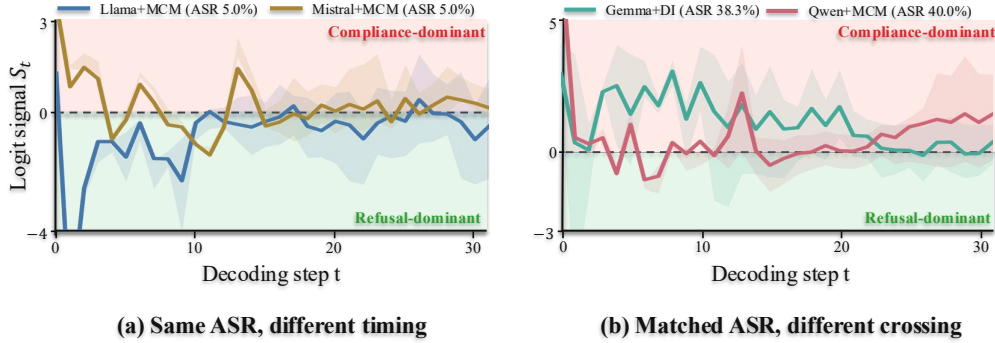


Figure 5: **ASR-neighboring conditions can have different raw trajectories.** Representative condition pairs with identical or near-identical ASR show different LMS crossing behavior. These raw trajectories are qualitative motivation only; the main text uses calibrated RP-plane distances for the primary comparison.

B Lexicon and Method Details

B.1 Lexicon Specification

The Logit-Margin Score uses a fixed English compliance–refusal lexicon to keep the diagnostic interpretable and training-free. The lexicon is intentionally small; Appendix F.5 and Appendix F.9 test whether this choice drives the conclusions.

Refusal Lexicon ($\mathcal{L}_{\text{ref}}$, 26 entries).

"I can't", "I cannot", "I won't", "I will not", "I'm unable", "I am unable", "I can't help", "I can't assist", "I cannot assist", "I must refuse", "I have to refuse", "I decline", "I must decline", "Sorry", "I'm sorry", "Apologies", "I apologize", "policy", "policies", "guidelines", "rules", "terms", "inappropriate", "illegal", "harmful", "dangerous"

Compliance Lexicon ($\mathcal{L}_{\text{cmp}}$, 21 entries).

"Sure", "Certainly", "Of course", "Here's", "Below is", "Step 1", "Step 2", "First", "Second", "Third", "Next", "Instructions",

"Guide", "Tutorial", "Procedure", "You can", "To do this", "Do the following", "Detailed", "Comprehensive", "Complete"

Implementation details. Surface phrases are expanded to tokenizer-level IDs, including bare and whitespace-prefixed forms, and duplicate IDs are removed. For multi-token surface phrases such as Step 1, we take the union of all constituent tokenizer IDs produced by these encodings rather than matching an ordered phrase at runtime. This order-free expansion can include common constituent tokens, such as digits or frequent subword pieces, that also appear in benign contexts. We treat this as a deliberate simplification rather than a phrase-level matcher: the analyses aggregate over compliance–refusal margins, calibrate those margins against matched harmful and attack-formatted benign references, and remain stable under lexicon perturbations and lexicon-internal top- k aggregation choices (Appendix F.5; Appendix F.6). These checks suggest that union-token artifacts are unlikely to explain the observed temporal separation, while phrase-level or order-aware matching remains a useful refinement for future implementations. At each decoding step t , logits are aggregated over each tokenized lexicon using top- k selection within the lexicon with $k = 10$:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad (4)$$

$$\mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \quad (5)$$

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (6)$$

We use $k = 10$ to smooth single-token volatility without averaging over the full tokenized lexicon; Appendix F.6 reports sensitivity to this choice.

B.2 Multi-Turn Context Manipulation Protocol

MCM represents multi-turn context-level pressure rather than token-level suffix optimization. The attack submits four turns: three context-normalizing turns followed by the target harmful request. Unlike single-prompt simulations of dialogue history, this protocol lets the model’s own previous responses become part of the next-turn context.

B.3 Benign Sample Exclusion Rationale

Sign reversal is computed only for harmful samples because benign sign patterns mainly reflect prompt format rather than safety activation. For example, MCM benign samples consistently show $S > 0$, while GCG benign samples often show negative S because their single-turn format differs structurally from MCM. We therefore use attack-formatted benign references for RP calibration and reserve sign-reversal interpretation for harmful prompts.

C RP-Plane Calibration and Coarse Annotations

RP calibration is reference-relative rather than a claim that raw LMS values are directly comparable. Values outside $[0, 1]$ indicate that successful attacks fall outside the harmful–benign reference interval for that condition. Negative RP values are reported as observed inversions rather than forced into a separate causal interpretation.

C.1 Reference-Anchor Sensitivity

The primary RP anchor uses all harmful samples in a condition as M_{harmful} . This choice avoids requiring failed jailbreak examples, but it also creates a known sensitivity: in high-ASR conditions, successful samples contribute heavily to the harmful anchor and can shrink the apparent displacement between M_{success} and M_{harmful} . For this reason, coarse RP-plane annotations should be read as observations under a fixed reference protocol, not as explanations of underlying causes. The issue is most relevant for high-ASR conditions such as Mistral+DI (80.0%) and Qwen+GCG (76.7%), and least relevant for low-ASR conditions where failures dominate the harmful anchor.

High-ASR anchor compression. For a fixed model–attack condition whose harmful anchor is computed from all harmful samples, the all-harmful reference can be written as

$$M_{\text{harmful}} = (1 - \text{ASR})M_{\text{failed}} + \text{ASR}M_{\text{success}}.$$

Thus,

$$M_{\text{success}} - M_{\text{harmful}} = (1 - \text{ASR})(M_{\text{success}} - M_{\text{failed}}).$$

As ASR increases, successful samples contribute more heavily to the all-harmful anchor and can compress the apparent displacement from the primary harmful reference. We therefore interpret NGB as no reliable RP-plane displacement under the fixed reference protocol, not as evidence that no safety-related signal exists anywhere. The main text therefore uses RP-plane coordinates and coarse annotations to expose model–attack heterogeneity under this fixed reference protocol. Alternative anchors answer different questions. This anchor dependence does not imply that one anchor is universally correct; rather, each anchor defines a different comparison question. A failed-only harmful anchor would isolate successful attacks from observed failures, but it is unstable in high-ASR conditions and unavailable when a condition has too few failed samples. A format-free benign anchor would estimate deployment-like benign behavior, but would remove the attack wrapper that RP is designed to control. We therefore use attack-formatted benign references for RP calibration and report format-free benign queries separately as a false-positive characterization.

C.2 RP-Plane Sensitivity Checks

RP-plane sensitivity checks test whether calibrated coordinates are driven by denominator instability, ASR-neighbor sampling noise, anchor choice, or raw-logit scale. All checks use the same 12-condition Llama-Guard primary grid as the main text. The primary filter reproduces the condition-level ASR, RP_A , RP_B , and r values in Table 1.

Denominator diagnostics. The RP denominator is stable in nearly all condition axes. Table 6 reports $\Delta_A = M_{\text{benign}}(S_0) - M_{\text{harmful}}(S_0)$ and $\Delta_B = M_{\text{benign}}(\bar{S}) - M_{\text{harmful}}(\bar{S})$. Only one condition-axis, Mistral+DI on Δ_A , falls below the near-zero threshold $|\Delta| < 0.10$.

Table 6: Denominator sign and magnitude audit for RP calibration. Values are benign-minus-harmful anchor gaps; “Flag” marks $|\Delta| < 0.10$.

Condition	$\Delta_A (S_0)$	$\Delta_B (\bar{S})$	Flag
Gemma + DI	2.06	0.60	–
Gemma + GCG	4.16	0.39	–
Gemma + MCM	5.06	1.84	–
Llama + DI	2.06	0.83	–
Llama + GCG	3.67	0.46	–
Llama + MCM	4.93	2.68	–
Mistral + DI	0.06	0.26	Δ_A
Mistral + GCG	1.62	−0.68	–
Mistral + MCM	0.74	0.98	–
Qwen + DI	4.65	0.45	–
Qwen + GCG	5.67	−1.32	–
Qwen + MCM	0.92	1.55	–

ASR-neighboring pair bootstrap. Pairwise RP-plane distances use stratified bootstrap resampling within each condition’s success, failure, and benign groups, holding the observed ASR and success count fixed. This provides descriptive CIs for D_{RP} , not a pairwise permutation null distribution.

Anchor sensitivity. Failed-only harmful anchors preserve raw axis dominance in 11 of 12 conditions, while model-global anchors change raw axis dominance in 5 of 12 conditions. This supports the condition-relative anchor as the primary coordinate system and treats model-global anchoring as a sensitivity analysis.

Table 7: ASR-neighboring condition pairs ($|\Delta \text{ASR}| \leq 5$ percentage points) with stratified-bootstrap 95% CIs for D_{RP} ($B = 10,000$).

Pair	ASR diff. (pp)	D_{RP} [95% CI]	n_{succ}
Llama+GCG vs. Llama+MCM	3.3	3.156 [2.028, 6.377]	5/3
Llama+GCG vs. Mistral+MCM	3.3	3.059 [1.926, 6.256]	5/3
Llama+DI vs. Mistral+GCG	1.7	1.014 [0.726, 1.351]	33/34
Gemma+DI vs. Qwen+MCM	1.7	0.625 [0.376, 1.131]	23/24
Llama+DI vs. Qwen+DI	3.3	0.572 [0.237, 1.026]	33/35
Gemma+DI vs. Gemma+GCG	5.0	0.508 [0.296, 1.041]	23/20
Mistral+GCG vs. Qwen+DI	1.7	0.508 [0.188, 0.885]	34/35
Llama+MCM vs. Mistral+MCM	0.0	0.330 [0.078, 0.851]	3/3
Mistral+DI vs. Qwen+GCG	3.3	0.184 [0.033, 1.868]	48/46
Median across nine pairs	–	0.572 [0.489, 0.924]	–

Table 8: Anchor sensitivity of RP coordinates. Each cell reports $(\text{RP}_A, \text{RP}_B)$ /dominant raw axis before applying NGB filtering.

Condition	Primary condition anchor	Failed-only harmful anchor	Model-global anchor
Gemma + DI	(0.72, 0.55)/A	(0.80, 0.66)/A	(0.40, 0.57)/B
Gemma + GCG	(0.27, 0.79)/B	(0.35, 0.85)/B	(0.26, 0.32)/B
Gemma + MCM	(0.09, 0.22)/B	(0.10, 0.26)/B	(0.14, 0.21)/B
Llama + DI	(0.35, 0.84)/B	(0.55, 0.92)/B	(0.84, 1.08)/B
Llama + GCG	(1.01, 3.80)/B	(1.01, 3.08)/B	(0.61, 1.21)/B
Llama + MCM	(0.79, 0.66)/A	(0.80, 0.67)/A	(0.90, 0.89)/A
Mistral + DI	(−0.18, 0.00)/A	(−3.10, 0.01)/A	(−0.66, 1.82)/B
Mistral + GCG	(0.30, −0.17)/A	(0.50, −0.51)/B	(0.08, 0.98)/B
Mistral + MCM	(0.49, 0.79)/B	(0.50, 0.80)/B	(1.62, 1.96)/B
Qwen + DI	(0.14, 0.31)/B	(0.28, 0.52)/B	(−0.13, 3.18)/B
Qwen + GCG	(0.01, 0.00)/A	(0.02, 0.00)/A	(−0.49, 0.53)/B
Qwen + MCM	(0.32, 0.07)/A	(0.43, 0.11)/A	(0.87, −2.63)/B

Rank-based RP. A rank-based RP variant replaces raw LMS scale with percentile positions within the harmful-plus-benign reference distribution for each condition. The rank-based coordinates strongly track the primary coordinates: Pearson $r = 0.858$ for RP_A , $r = 0.920$ for RP_B , and $r = 0.925$ for displacement magnitude.

Table 9: Rank-based RP sensitivity check. Rank RP uses empirical percentile positions rather than raw LMS scale.

Condition	Primary (RP_A, RP_B)	Rank-based (RP_A, RP_B)	r_{rank}
Gemma + DI	(0.72, 0.55)	(0.24, 0.33)	0.41
Gemma + GCG	(0.27, 0.79)	(0.20, 0.72)	0.75
Gemma + MCM	(0.09, 0.22)	(0.16, 0.29)	0.33
Llama + DI	(0.35, 0.84)	(0.07, 1.23)	1.24
Llama + GCG	(1.01, 3.80)	(0.99, 2.01)	2.24
Llama + MCM	(0.79, 0.66)	(0.49, 0.45)	0.67
Mistral + DI	(−0.18, 0.00)	(−0.19, 0.00)	0.19
Mistral + GCG	(0.30, −0.17)	(0.25, −0.16)	0.30
Mistral + MCM	(0.49, 0.79)	(0.66, 0.75)	1.00
Qwen + DI	(0.14, 0.31)	(0.06, 0.09)	0.11
Qwen + GCG	(0.01, 0.00)	(0.01, −0.01)	0.01
Qwen + MCM	(0.32, 0.07)	(0.20, 0.06)	0.21

C.3 Permutation Test for Null-generation Bias

The success-label permutation test decides whether a calibrated displacement is reliably nonzero before assigning a coarse RP-plane annotation. Within each model–attack condition, we permute success/failure labels, recompute $r = \sqrt{\text{RP}_A^2 + \text{RP}_B^2}$, and estimate one-sided permutation p -values with $B = 10,000$ permutations. The primary family contains the 12 Llama-Guard model–attack conditions, and Benjamini–Hochberg FDR correction is applied at $q = 0.05$. Conditions that do not survive correction receive the Null-generation Bias annotation, meaning the calibrated trajectory

summaries provide no reliable evidence for either pre-generation or in-generation bias. The rerun preserves seven reliable RP-displacement conditions, corresponding to the non-NGB cells in Table 1.

Table 10: Permutation test rerun for calibrated displacement r . The PGB/IGB/NGB annotations use FDR-corrected significance followed by axis dominance.

Model + Attack	r_{obs}	p_{perm}	p_{BH}	FDR result	Annotation status
Gemma + DI	0.90	<.001	<.001	Significant	Axis-dominant
Gemma + GCG	0.83	.002	.004	Significant	Axis-dominant
Gemma + MCM	0.24	<.001	.002	Significant	Axis-dominant
Llama + DI	0.91	<.001	<.001	Significant	Axis-dominant
Llama + GCG	3.94	<.001	<.001	Significant	Axis-dominant
Llama + MCM	1.03	.001	.002	Significant	Axis-dominant
Mistral + DI	0.18	.500	.546	Not significant	Null-generation Bias
Mistral + GCG	0.35	<.001	<.001	Significant	Axis-dominant
Mistral + MCM	0.93	.054	.081	Not significant	Null-generation Bias
Qwen + DI	0.34	.070	.093	Not significant	Null-generation Bias
Qwen + GCG	0.01	.988	.988	Not significant	Null-generation Bias
Qwen + MCM	0.32	.280	.336	Not significant	Null-generation Bias

Bootstrap intervals provide a descriptive stability check for calibrated displacement magnitudes. They complement the primary permutation test but do not override its FDR-corrected annotation decision. Intervals for very small n_{succ} are descriptive only and are not used for annotation assignment.

Table 11: Bootstrap confidence intervals for selected calibrated displacement magnitudes r . PGB/IGB/NGB annotations follow the primary permutation test and axis-dominance rule.

Model + Attack	n_{succ}	r	95% CI	Annotation
Gemma + DI	23	0.90	[0.73, 1.08]	PGB
Gemma + GCG	20	0.83	[0.34, 1.41]	IGB
Gemma + MCM	11	0.24	[0.12, 0.33]	IGB
Llama + DI	33	0.91	[0.55, 1.21]	IGB
Llama + GCG	5	3.94	[3.05, 5.32]	IGB
Llama + MCM	3	1.03	[0.95, 1.09]	PGB
Mistral + DI	48	0.18	[0.07, 1.37]	NGB
Mistral + GCG	34	0.35	[0.17, 0.57]	PGB [‡]
Mistral + MCM	3	0.93	[0.44, 1.43]	NGB [§]
Qwen + DI	35	0.34	[0.09, 0.68]	NGB [¶]
Qwen + GCG	46	0.01	[0.02, 0.22]	NGB [†]
Qwen + MCM	24	0.32	[0.05, 0.81]	NGB

[†]Qwen+GCG has a near-zero point estimate and non-significant permutation test, but its bootstrap interval is unstable; we therefore interpret it as no reliable RP-axis displacement, not as a proven absence of all safety evidence. [‡]Mistral+GCG has an inverted generation-time RP value. [§]Mistral+MCM shows large displacement magnitude but does not pass the primary permutation test. [¶]Qwen+DI is borderline under the raw permutation test and does not survive the primary significance stage.

C.4 Axis-Dominance and Small-Sample Caveats

Coarse RP-plane annotations are compact summaries of axis structure, not independently established causes. They are most fragile when the number of successful attacks is small or when the axis-dominance margin between $|\text{RP}_A|$ and $|\text{RP}_B|$ is close. After the significance stage, axis dominance is assigned by the larger absolute RP coordinate:

$$\text{PGB} : |\text{RP}_A| > |\text{RP}_B|, \quad \text{IGB} : |\text{RP}_A| < |\text{RP}_B|. \quad (7)$$

Axis-indeterminate exact ties are reported as N/A. Table 12 reports the raw axis margin used after the significance stage.

The most cautious reading is therefore condition-level and observational: PGB/IGB annotations indicate which calibrated axis dominates under the primary pipeline, while NGB means no reliable RP-axis displacement after the permutation/FDR stage.

Table 12: Axis-dominance margins in the 12-condition grid. Positive margin favors PGB; negative margin favors IGB. NGB annotations override axis dominance when the permutation test is not significant.

Model	Attack	n_{succ}	RP_A	RP_B	$ \text{RP}_A - \text{RP}_B $	Reading
Gemma	DI	23	0.72	0.55	0.17	PGB
Gemma	GCG	20	0.27	0.79	-0.52	IGB
Gemma	MCM	11	0.09	0.22	-0.13	IGB, small margin
Llama	DI	33	0.35	0.84	-0.49	IGB
Llama	GCG	5	1.01	3.80	-2.79	IGB, low n_{succ}
Llama	MCM	3	0.79	0.66	0.14	PGB, low n_{succ}
Mistral	DI	48	-0.18	0.00	0.18	NGB by permutation
Mistral	GCG	34	0.30	-0.17	0.13	PGB, inverted/small margin
Mistral	MCM	3	0.49	0.79	-0.30	NGB by permutation, low n_{succ}
Qwen	DI	35	0.14	0.31	-0.17	NGB by permutation, borderline
Qwen	GCG	46	0.01	0.00	0.00	NGB
Qwen	MCM	24	0.32	0.07	0.25	NGB by permutation

C.5 Condition-Level RP-Plane Annotations and Behavioral Probe Response

Table 1 in the main text reports the per-condition RP-plane coordinates, coarse PGB/IGB/NGB annotations, and behavioral-probe response. The annotation column is an observational summary of calibrated logit evidence, not a causal explanation. The behavioral-probe columns are included only to show whether the coarse annotations correspond to different behavioral responses.

C.6 Entropy Check for Null-generation Bias

Entropy provides an independent check for whether a Null-generation Bias annotation is merely a lexicon artifact. For Qwen+GCG, generation-time entropy shows almost no differentiation between successful attacks and the harmful baseline ($|d| = 0.077$), matching the near-zero calibrated LMS displacement. For contrast, Llama+MCM shows measurable entropy differentiation ($|d| = 0.103$ – 0.416), showing that entropy can detect condition-level variation when such variation is present.

Table 13: Entropy cross-validation for selected conditions.

Condition	ASR	Entropy $ d $	Lexicon $ d $	Reading
Qwen + GCG	76.7%	0.077	0.005–0.011	No detectable entropy/LMS displacement
Llama + MCM	5.0%	0.103–0.416	0.217–0.256	Weak-to-medium displacement

C.7 Intervention-Relevance Implications

The coarse RP-plane annotations suggest different intervention-facing readings without claiming hidden causes. IGB conditions are the most natural fit for generation-time monitors because the discriminative signal appears during decoding. PGB conditions instead suggest that calibrated evidence is already present at the pre-generation position. NGB conditions should be treated as observability boundaries: the current calibrated trajectory summaries do not provide reliable evidence for either axis.

D Temporal Evidence Supporting Success Prediction

D.1 Token Rank Linkage

Token-rank analysis verifies that LMS reflects actual next-token competition. When $S_t < 0$, the highest-ranked refusal token typically appears near the top of the vocabulary; when $S_t > 0$, refusal tokens drop by orders of magnitude. Successful jailbreaks concentrate in compliance-dominant regions where refusal tokens remain low-ranked, supporting the main-text interpretation that delayed or absent refusal dominance is visible in the S_t trajectory.

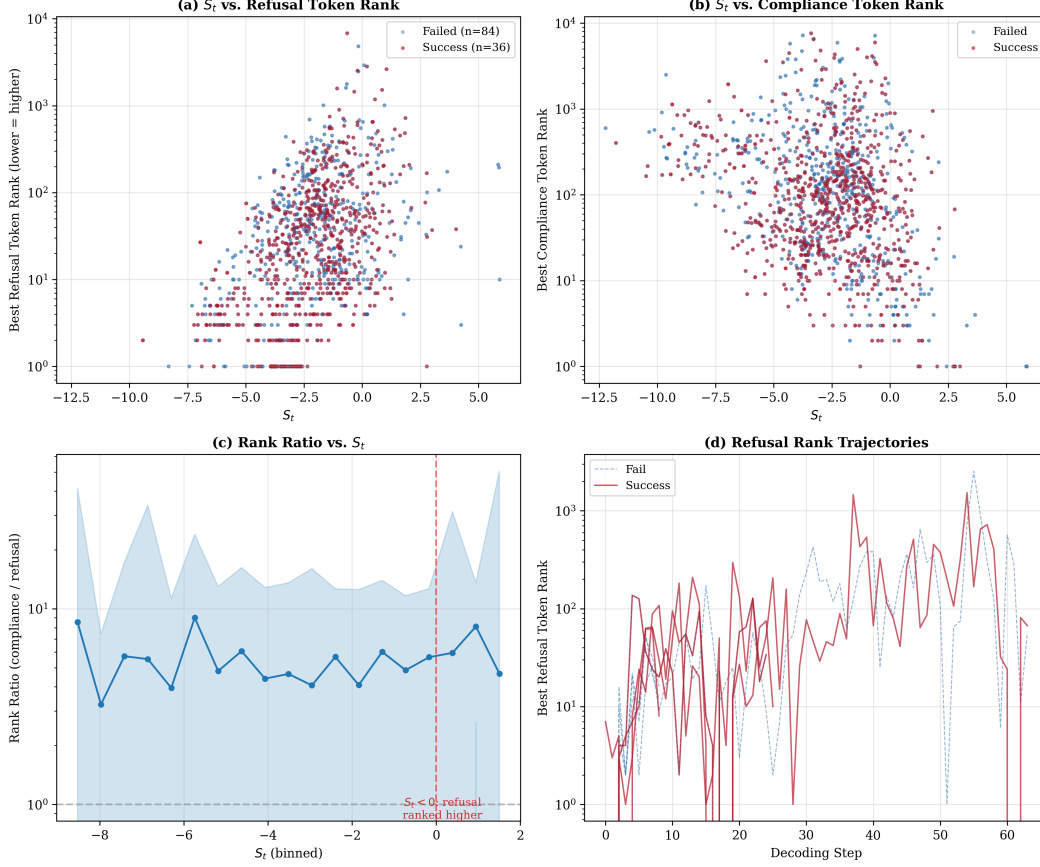


Figure 6: **Token rank linkage.** Negative S_t corresponds to high-ranked refusal tokens, while positive S_t pushes refusal tokens lower in the vocabulary ranking.

E Hidden-State References

Hidden-state references define the access–observability boundary of the S_t trajectory. Hidden-state probes use internal activations, so their higher predictive capacity is expected under stronger access assumptions. The relevant question is where the lower-access S_t trajectory preserves temporally structured evidence and where it weakens.

E.1 Direction Construction

The refusal-direction projection uses a per-model, stronger-access hidden-state construct-validity reference rather than an explanatory label. For each model m , we fix the closest layer to 50% network depth before any alignment evaluation (L50; Llama/Mistral layer 16, Qwen layer 14, Gemma layer 21). This gives a pre-specified midpoint reference rather than an outcome-tuned hidden-state probe. At this fixed layer, we compute a difference-of-means direction from paired harmful and benign reference prompts at the final prompt token:

$$\mathbf{d}_m = \frac{\bar{\mathbf{h}}_{\text{harmful},m} - \bar{\mathbf{h}}_{\text{benign},m}}{\|\bar{\mathbf{h}}_{\text{harmful},m} - \bar{\mathbf{h}}_{\text{benign},m}\|_2}. \quad (8)$$

The direction uses prompt-reference category rather than jailbreak success labels, and is then held fixed while each decoding-time hidden state is projected as $z_t = \langle \mathbf{h}_t, \mathbf{d}_m \rangle$. In high-ASR conditions, this reference can also encode harmful-versus-benign context, so we treat it as a construct-validity reference rather than a pure refusal axis. Because higher S_t means compliance-dominant logits while higher z_t means movement toward the harmful/refusal reference direction, the expected temporal association is negative.

Table 14: **Condition-level temporal alignment between the S_t trajectory and refusal-direction projections.** Attack rows use the same order within each model: MCM, GCG, and DI.

Model	Attack	Temporal association			Interpretation	
		$ \bar{\rho} _{0:4}$	$\bar{\rho}_{0:4}$	$\bar{\rho}_{\text{lag}}$	Agree.	Reading
Llama	MCM	0.698	-0.698	0.599	0.758	Strong alignment
	GCG	0.780	-0.780	0.607	0.719	Strong alignment
	DI	0.794	-0.794	0.298	0.655	Strong alignment
Qwen	MCM	0.330	0.330	0.194	0.532	<i>Boundary</i>
	GCG	0.312	-0.312	0.227	0.462	Weak alignment
	DI	0.268	0.167	0.214	0.504	<i>Boundary</i>
Mistral	MCM	0.177	-0.174	0.196	0.531	Weak alignment
	GCG	0.460	-0.460	0.272	0.572	Moderate alignment
	DI	0.196	-0.137	0.208	0.579	Weak alignment
Gemma	MCM	0.217	-0.209	0.184	0.551	Weak alignment
	GCG	0.384	-0.384	0.241	0.615	Moderate alignment
	DI	0.501	-0.304	0.207	0.624	Moderate alignment

$|\bar{\rho}|_{0:4}$ measures early-step association strength as the mean absolute correlation between S_t and z_t over steps 0–4, $\bar{\rho}_{0:4}$ preserves its direction, $\bar{\rho}_{\text{lag}}$ measures within-trajectory similarity after lag optimization, and Agree. measures state-sign agreement.

E.2 Layer-Sweep Sensitivity

The fixed L50 layer is not the only layer at which the temporal alignment appears. To check whether the hidden-state reference result depends on the midpoint layer choice, we repeat the direction-construction and temporal-alignment analysis at three pre-specified relative depths: L25, L50, and L75. For each model and layer, the direction is recomputed from the same harmful-versus-benign reference categories at that layer and then held fixed while decoding-time hidden states are projected. We then summarize the harmful attack conditions (MCM, GCG, DI) using the same early-step association statistic, $|\bar{\rho}|_{0:4}$.

The qualitative pattern is stable across layers. Llama remains strongly aligned at all three depths, Qwen remains the main boundary case, and Mistral/Gemma retain attack-dependent moderate alignment rather than collapsing at non-L50 layers. The L50 protocol is therefore a fixed midpoint reference, not a layer chosen to maximize the reported alignment.

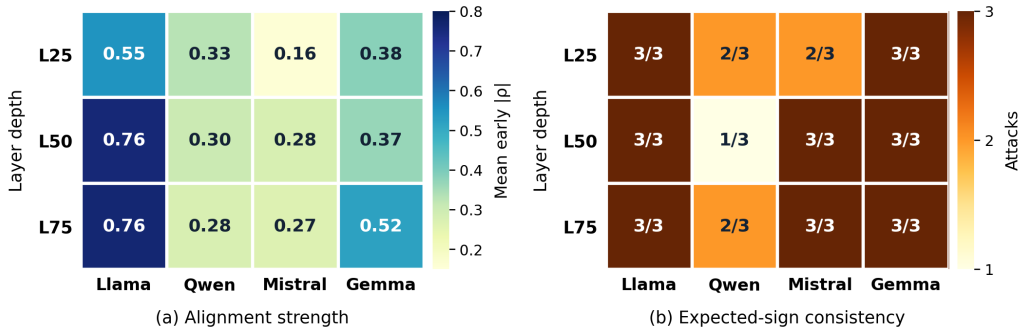


Figure 7: **Layer-sweep hidden-state alignment.** Panel (a) shows mean early $|\rho|$ between S_t and z_t ; panel (b) shows expected-sign consistency across attacks. Llama is consistently strongest, while Qwen is the main boundary case.

The table shows that both S_t summaries and hidden-state probes carry above-chance success-label information across the model grid. Hidden-state probes generally have higher mean AUC under stronger access assumptions, while S_t summaries remain informative without activation access. This supports the main claim of the Temporal Logit Observability (TLO) protocol: it preserves an accessible subset of safety-related temporal evidence rather than replacing hidden-state probes.

Table 15: **Hidden-state layer-sweep sensitivity.** Entries report the mean $|\overline{\rho}|_{0:4}$ across the three harmful attack conditions at each pre-specified relative depth. Parentheses report the number of harmful conditions with the expected negative signed association.

Model	L25	L50	L75
Llama	0.548 (3/3)	0.757 (3/3)	0.763 (3/3)
Qwen	0.330 (2/3)	0.303 (1/3)	0.275 (2/3)
Mistral	0.159 (2/3)	0.278 (3/3)	0.266 (3/3)
Gemma	0.382 (3/3)	0.367 (3/3)	0.520 (3/3)

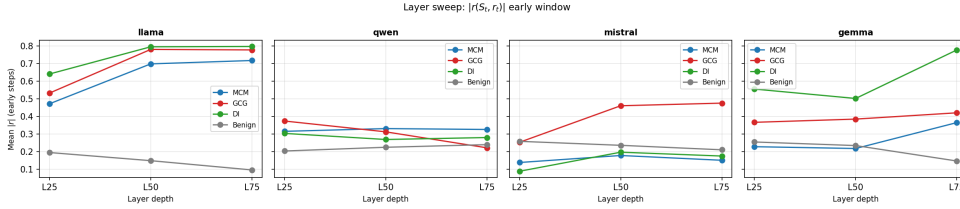


Figure 8: **Layer-sweep visualization for hidden-state alignment.** The plot shows early temporal association between the S_t trajectory and hidden-state reference projections across L25, L50, and L75. The same broad interpretation holds away from L50: alignment is strongest and most consistent for Llama, weaker and boundary-like for Qwen, and condition-dependent for Mistral and Gemma.

F Robustness and Ablations

F.1 Stochastic Decoding

Stochastic decoding preserves the main temporal structure at the prompt level. The auxiliary stochastic suite uses Llama stress conditions: Base, DecoyA (prefix decoy), DecoyB (token decoy), and PivotC (multi-turn pivot), with 10 samples per prompt across temperature $\tau \in \{0.0, 0.3, 0.7, 1.0\}$ and nucleus sampling $p \in \{0.9, 0.95, 1.0\}$. We aggregate repeated samples per prompt before computing reliability, avoiding trial-level pseudo-replication. DecoyB produced no LLM-judge successes in the companion AUC analysis, so the compact ICC summary below reports the three nonzero-success conditions.

F.2 Activation Timing Sensitivity

The first-crossing definition is not an artifact of a one-step fluctuation. Generation trajectories can oscillate around zero after the initial transition, so we test a stricter *confirmed crossing* variant that requires n consecutive refusal-dominant steps:

$$t_{\text{cross}}^{(n)} = \min\{t \geq 1 : S_t < 0, \dots, S_{t+n-1} < 0\}, \quad (9)$$

with right-censoring at T when no confirmed crossing occurs. The default metric is $t_{\text{cross}}^{(1)}$. On the current 12-condition Llama-Guard grid, the $n = 2$ criterion slightly increases both AUC and effect size, while $n = 3$ remains significant but becomes more conservative because short or oscillatory trajectories are censored.

The same qualitative pattern holds under stricter temporal criteria. Strong in-generation conditions retain high temporal discriminability, while Qwen+GCG remains near chance, supporting the Null-generation Bias interpretation as an observability boundary rather than a threshold artifact.

F.3 Learned Baseline Refresh

The trained-classifier check asks whether a fitted classifier can extract substantially more success-label information from the same S_t trajectory features. On the 12-condition Llama-Guard grid, logistic regression improves AUC modestly but does not replace the fixed temporal diagnostics. The 5-feature set is $\{S_0, \bar{S}, \bar{S}_{1:1}, \bar{S}_{1:3}, \bar{S}_{1:5}\}$; the 9-feature set adds $\min_t S_t$, $\max_t S_t$, $\text{std}_t(S_t)$, and range; the 13-feature set further adds positive-step rate, t_{cross} , t_{cross}/T , and sign flips.

Table 16: Success-prediction reference values for S_t summaries and hidden-state probes. Hidden-state probes use stronger access assumptions and serve as construct-validity references; S_t summaries report hidden-state-free logit information.

Family	Method	Llama	Qwen	Mistral	Gemma	Avg
S_t summaries	S_0	0.852	0.555	0.647	0.738	0.698
	$\bar{S}_{1:1}$	0.853	0.556	0.648	0.739	0.699
	$\bar{S}_{1:3}$	0.874	0.570	0.667	0.667	0.694
	$\bar{S}_{1:5}$	0.883	0.574	0.634	0.683	0.694
	$\bar{S}$	0.943	0.551	0.684	0.796	0.744
	sign flips	0.894	0.572	0.772	0.679	0.729
Hidden dir.	direction, full	0.874	0.728	0.905	0.797	0.826
	direction, $k = 1$	0.862	0.573	0.699	0.765	0.725
	direction, $k = 3$	0.865	0.598	0.756	0.785	0.751
	direction, $k = 5$	0.861	0.588	0.770	0.793	0.753
Hidden LogReg	logreg, full	0.779	0.733	0.842	0.804	0.790
	logreg, $k = 1$	0.866	0.717	0.693	0.782	0.765
	logreg, $k = 3$	0.800	0.751	0.723	0.843	0.779
	logreg, $k = 5$	0.855	0.730	0.796	0.829	0.803

Table 17: Prompt-level stochastic decoding robustness for trajectory metrics. ICC values summarize stability across repeated stochastic samples; S_0 is deterministic for a fixed prompt and is excluded from this reliability table.

Metric	Base	DecoyA	PivotC
t_{cross}/T ICC	0.944	0.927	0.919
Sign-flip ICC	0.899	0.919	0.925

The fitted 13-feature classifier gains 4.2 pp over fixed t_{cross} and 5.5 pp over fixed $\bar{S}_{1:5}$. This shows that additional trajectory features contain modest complementary information, but the primary contribution remains the interpretable temporal diagnostic rather than maximum-AUC success prediction.

F.4 Non-Lexical Baseline

Non-lexical controls show that generic logit statistics do not explain away the compliance–refusal margin. On a controlled Llama subset, entropy is informative and logit norm is near random, while the margin keeps the interpretation tied to refusal–compliance competition.

F.5 Lexicon Perturbation

The observed signal is stable under reasonable lexicon perturbations. Because this controlled Llama subset contains few successful attacks, we treat it as a sanity check rather than primary robustness evidence. All tested variants retain similar discrimination on this subset, indicating that the result is not driven by a single refusal or compliance token. We test five variants on Llama-3.1-8B-Instruct ($n = 60$): **original**, **no_sorry**, **extended_refusal**, **extended_compliance**, and **minimal** (5-entry: Sorry, illegal, harmful, Sure, Step 1).

Key findings. On this subset, all variants achieve $\text{AUC} \geq 0.964$ ($\max |\Delta\text{AUC}| = 0.018$). Even this 5-entry minimal lexicon maintains similar discrimination ($\text{AUC} = 0.978$), supporting the sanity check while leaving the primary evidence to the pooled grid. Figure 9 visualizes the score distributions across variants.

F.6 Lexicon Aggregation Top- k Sensitivity

The default aggregation size $k = 10$ balances two effects: smaller values are more sensitive to single-token spikes, while larger values increasingly average over lower-ranked lexicon tokens. On

Table 18: t_{cross} sensitivity to stricter confirmed-crossing thresholds. Global rows use the 12-condition harmful grid ($n = 720$); selected per-condition AUC rows show representative stable and boundary cases.

	$n = 1$	$n = 2$	$n = 3$
<i>Global, all conditions pooled</i>			
AUC	0.784	0.791	0.770
Cohen’s d	0.796	0.830	0.440
p -value	$< 10^{-20}$	$< 10^{-20}$	$< 10^{-7}$
<i>Per-condition AUC, selected</i>			
Llama + DI	0.799	0.809	0.836
Llama + GCG	0.987	0.938	0.925
Gemma + DI	0.792	0.828	0.862
Gemma + GCG	0.629	0.719	0.651
Qwen + DI	0.653	0.675	0.652
Qwen + GCG	0.519	0.502	0.560

Table 19: Learned baseline comparison on the 12-condition grid ($n = 720$ harmful samples; 5-fold stratified cross-validation for LogReg rows). Fixed metrics require no training.

Method	# Features	AUC	95% CI
<i>Fixed metrics</i>			
t_{cross}	1	0.784	[0.750, 0.816]
$\tilde{S}_{1:5}$	1	0.771	[0.738, 0.804]
$\tilde{S}$	1	0.758	[0.723, 0.792]
S_0	1	0.680	[0.641, 0.718]
<i>Logistic regression</i>			
LogReg($\tilde{S}_{1:5}$)	1	0.769	[0.734, 0.802]
LogReg($S_0 + \tilde{S}$)	2	0.763	[0.728, 0.797]
LogReg(5 temporal)	5	0.813	[0.781, 0.844]
LogReg(9 trajectory)	9	0.806	[0.774, 0.837]
LogReg(13 features)	13	0.826	[0.795, 0.856]

the Llama+MCM harmful subset, changing k alters absolute scale but preserves the refusal-dominant generation-time interpretation.

F.7 Metric Decomposition

A natural concern is whether the compliance lexicon contributes genuine discriminative signal or merely reflects response-onset artifacts such as “Sure” and “Certainly” appearing at the beginning of any response. We therefore decompose $S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t)$ into three variants: S_{margin} (the full margin used in the paper), $S_{\text{ref}} = -\mu_{\text{ref}}$ (refusal-only), and $S_{\text{cmp}} = \mu_{\text{cmp}}$ (compliance-only). The margin form matters because it removes common logit movement shared by compliance and refusal lexicons. Together with the lexicon perturbation check above, this decomposition tests whether the diagnostic is merely tracking stylistic response-openers rather than safety-relevant compliance–refusal competition.

Common-mode rejection. The two component means are highly correlated ($r = 0.979$), sharing a dominant common factor that reflects per-step logit scale rather than safety-specific evidence. The shared component $(\mu_{\text{ref}} + \mu_{\text{cmp}})/2$ achieves only AUC = 0.594 in the ablation subset, while the differential score reaches 0.787. This confirms that the margin isolates safety-relevant variation by canceling shared movement, analogous to differential signal processing. The normalization is not merely a per-model correction: even after within-model z-scoring, the margin retains an +0.088 AUC advantage over refusal-only for generation-mean summaries.

Compliance is not merely an onset artifact. The margin outperforms refusal-only in most per-condition comparisons (Table 24), with the largest gains in conditions where compliance variation is outcome-dependent. For example, in Gemma conditions, μ_{cmp} can differ sharply between successes

Table 20: Non-lexical control on a controlled Llama subset ($n = 60$).

Metric	AUC	Feature group
$\bar{S}$	0.932	Logit margin
Compliance LogP	0.932	Lexicon component
Refusal LogP	0.915	Lexicon component
Entropy	0.898	Non-lexical
Logit Norm	0.441	Non-lexical

Table 21: Lexicon perturbation results with LLM-judge success labels ($n = 60$).

Variant	$S_0 (\mu \pm \sigma)$	$\bar{S} (\mu \pm \sigma)$	$AUC(S_0)$	$AUC(\bar{S})$	ΔAUC
original	$+1.55 \pm 1.95$	-1.44 ± 0.73	0.967	0.971	—
no_sorry	$+2.21 \pm 1.78$	-1.39 ± 0.73	0.960	0.971	+0.000
extended_refusal	-0.43 ± 2.07	-1.85 ± 0.67	0.964	0.989	+0.018
extended_compliance	$+4.00 \pm 1.95$	$+0.52 \pm 0.55$	0.964	0.964	-0.007
minimal	-2.61 ± 1.74	-1.57 ± 0.49	0.971	0.978	+0.007

and failures, so the compliance side adds signal rather than only marking generic response starts. Refusal-only and compliance-only variants can be informative in individual conditions, but the full margin is the most stable cross-condition summary.

F.8 Data-Driven Lexicon Induction

The handcrafted lexicon used throughout this paper was designed for interpretability and cross-model portability. A natural concern is whether results depend critically on specific lexicon choices, and whether a data-driven alternative would yield substantially different conclusions. We address this through log-odds-based lexicon induction and cross-tokenizer analysis.

Cross-tokenizer analysis. We first verify that the handcrafted lexicon remains compact and functional across all four model tokenizers. Table 25 reports the number of token IDs produced by expanding each lexicon phrase through each model’s tokenizer, including bare and whitespace-prefixed forms.

The lexicon occupies less than 0.23% of each vocabulary, yet achieves consistent AUC across all four models. Mistral’s smaller vocabulary produces fewer token IDs due to coarser BPE segmentation, but this does not degrade performance.

Induction method. We induce data-driven lexicons via word-level log-odds ratios. Using LLM-as-judge labels (`is_success_llm`) from Llama-Guard-3-8B, we partition response texts into compliant and refusal-like outputs. The compliant set contains harmful prompts where the attack succeeded; the refusal set contains failed harmful prompts plus benign responses. For each word w appearing at least five times, we compute

$$\text{LOR}(w) = \log P(w \mid \text{compliant}) - \log P(w \mid \text{refusal})$$

with Laplace smoothing ($\alpha = 1$). The top-25 words by $|\text{LOR}|$ form the induced refusal and compliance lexicons.

Semantic direction. The induced lexicons capture semantically meaningful but more content-specific patterns. Induced refusal words include benign-topic words such as *energy*, *sleep*, *cells*, *immune*, *ingredients*, *memory*, *cooking*, and *solar*, as well as safety markers such as *cannot* and *harmful*. Induced compliance words include domain-specific harmful-content terms such as *phishing*, *vpn*, *fetish*, *stealth*, *evade*, *rootkit*, *deepfake*, and *scam*. These differ qualitatively from the handcrafted lexicon, which targets generic safety behavior (*Sorry*, *cannot*, *Sure*, *Step 1*). The Jaccard overlap between induced and handcrafted word sets is near zero (refusal: 0.039, compliance: 0.000 for Llama-only; 0.000/0.000 for the full dataset), confirming that the two approaches capture largely complementary signals.

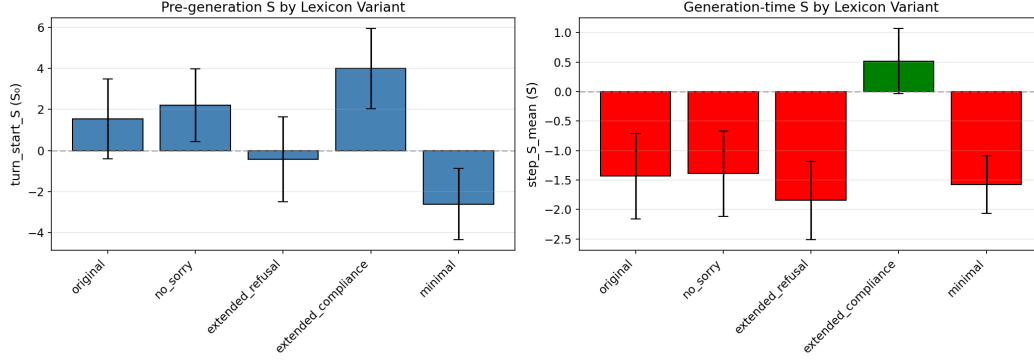


Figure 9: **Logit-Margin Score distributions by lexicon variant.** (Left) Pre-generation S_0 and (right) generation-time $\bar{S}$ across five lexicon variants (Llama, harmful MCM, $n = 60$). Absolute scale shifts with lexicon composition, but discriminatory structure is preserved across all variants; even `extended_compliance`, which shifts the baseline upward, maintains high AUC.

Table 22: Effect of lexicon aggregation size k on LMS summaries (Llama+MCM harmful subset; mean $\pm$ standard deviation, $n = 60$).

k	S_0	$\bar{S}$	Positive-step rate
5	$+0.57 \pm 2.57$	-1.21 ± 0.85	0.36 ± 0.13
10	$+1.45 \pm 2.02$	-1.42 ± 0.73	0.27 ± 0.12
25	$+1.40 \pm 1.55$	-1.47 ± 0.53	0.15 ± 0.09
50	$+0.30 \pm 1.05$	-1.09 ± 0.34	0.12 ± 0.08

AUC comparison. Table 26 compares three approaches: real-time logit-based S_t metrics, a bag-of-words (BoW) proxy using the handcrafted lexicon, and a BoW proxy using the induced lexicon. The BoW proxy computes $S_{\text{BoW}} = (n_{\text{refusal}} - n_{\text{compliance}}) / (n_{\text{refusal}} + n_{\text{compliance}})$ from word counts in the full response text. This diagnostic pool includes the primary grid plus auxiliary non-stochastic generations used for lexicon checks, so its sample sizes differ from the 12-condition primary grid.

Interpretation and caveats. The induced BoW lexicon achieves substantially higher AUC (0.936–0.996) than both the handcrafted BoW proxy (0.674–0.774) and the logit-based metrics (0.630–0.787). However, this comparison is not apples-to-apples: BoW operates on the complete response text after generation, while S_t is computed online from token-level logits. The induced BoW AUC should therefore be interpreted as an upper bound on lexicon quality, not as a direct competitor to real-time logit-level observability. Two conclusions follow. First, the handcrafted lexicon is a conservative lower bound: a data-driven lexicon can capture more success-label information, so the reported LMS AUC values do not exhaust the information available in token distributions. Second, the near-zero Jaccard overlap shows that handcrafted and induced lexicons capture complementary signals: generic refusal/compliance behavior versus domain-specific harmful content.

Per-attack induced lexicons. Table 27 shows how induced compliance words differ by attack paradigm, confirming that each attack elicits distinct harmful-content domains.

Notably, the handcrafted BoW proxy is near random for DI (0.500), indicating that DI-generated responses often avoid standard refusal keywords. The induced lexicon attains strong discrimination (0.971) by capturing domain-specific content. This per-attack variation reinforces the design rationale for the handcrafted lexicon: it sacrifices per-condition optimality for cross-condition consistency.

Transferability analysis. A key concern with data-driven lexicons is whether they generalize beyond the training distribution. We evaluate cross-attack and cross-model transferability by training the induced lexicon on one slice and testing it on others.

Induced lexicons transfer moderately across conditions (mean off-diagonal AUC ≈ 0.65) but substantially underperform their in-distribution AUC (≈ 0.93). This confirms that induced lexicons

Table 23: **Metric decomposition: AUC comparison.** The primary 12-condition summary uses the current Llama-Guard grid ($n = 720$). The detailed temporal rows report the 11-condition ablation subset used for component-wise diagnostics, where all three component metrics were available. Z-scored rows remove inter-model baseline shifts.

Metric	S_{margin}	$S_{\text{ref}} (-\mu_{\text{ref}})$	$S_{\text{cmp}} (\mu_{\text{cmp}})$	$\Delta_{\text{ref}-\text{margin}}$
<i>Primary 12-condition grid</i>				
Primary temporal summary	0.784	0.622	0.518	-0.162
<i>Raw AUC, component-wise ablation subset</i>				
S_0 (pre-generation)	0.598	0.580	0.505	-0.017
$\bar{S}$ (generation mean)	0.787	0.622	0.518	-0.165
$\bar{S}_{1:5}$	0.719	0.619	0.565	-0.100
t_{cross}	0.785	0.500	—	-0.285
<i>Z-scored within-model AUC, component-wise ablation subset</i>				
$\bar{S}$	0.806	0.718	0.545	-0.088
$\bar{S}_{1:5}$	0.716	0.659	0.586	-0.057

Table 24: **Per-condition $\bar{S}_{1:5}$ AUC by metric variant.** Component-wise ablation subset. Bold indicates the best metric per condition.

Condition	n	ASR	S_{margin}	S_{ref}	S_{cmp}
Gemma+GCG	60	33.3%	0.693	0.511	0.667
Gemma+MCM	60	23.3%	0.739	0.522	0.616
Llama+DI	60	55.0%	0.750	0.767	0.730
Llama+GCG	60	8.3%	0.964	0.935	0.938
Llama+MCM	60	11.7%	0.677	0.663	0.612
Mistral+DI	60	80.0%	0.634	0.526	0.741
Mistral+GCG	60	56.7%	0.827	0.761	0.719
Mistral+MCM	60	5.0%	0.708	0.684	0.643
Qwen+DI	60	58.3%	0.627	0.602	0.639
Qwen+GCG	60	76.7%	0.550	0.528	0.618

capture condition-specific vocabulary rather than universal safety behavior. The refusal word overlap across attack-specific lexicons is modest (Jaccard = 0.11–0.25 for refusal words; 0.02–0.04 for compliance words), explaining the transfer gap. Common refusal words across all conditions are often benign topic indicators rather than safety markers, while compliance words are almost entirely condition-specific.

Compatibility with temporal decomposition. The paper’s central diagnostic relies on the temporal structure of the S_t trajectory, specifically the decomposition into pre-generation (S_0) and generation-time ($\bar{S}$) components. This decomposition is fundamentally unavailable to post-hoc text classifiers, including induced BoW proxies. The induced BoW score computes a single scalar from the complete response text, collapsing all temporal information into one value. Consequently, RP_A cannot be computed because it requires S_0 , RP_B cannot be computed because it requires comparing S_0 against $\bar{S}$, sign reversal cannot be measured as a pre-generation-to-generation transition, and early- k detection curves cannot be evaluated without per-token logit access. In short, induced lexicons support the information content of safety-relevant vocabulary, but they cannot replace the temporal decomposition that makes LMS informative. The two approaches are complementary: logit-based S_t reveals when and how safety signals emerge, while data-driven lexicon induction confirms that vocabulary choice is not the limiting source of signal.

Surface-token and tokenizer audit scope. A remaining concern is that LMS may partially reflect surface-token or tokenizer artifacts rather than only a distributed compliance–refusal contrast. This concern is most relevant for multi-token surface phrases whose tokenizer expansion can introduce frequent subword pieces or digit tokens. Our lexicon perturbation, metric-decomposition, and data-driven lexicon checks reduce this concern, but they do not eliminate the need for finer-grained tokenizer audits. Future audits should report token-contribution concentration, leave-one-token-

Table 25: Cross-tokenizer expansion of the handcrafted lexicon. The same 47 lexicon phrases (26 refusal + 21 compliance) expand to 74–102 unique token IDs depending on tokenizer granularity.

Model	Vocab Size	Ref IDs	Cmp IDs	Lexicon %
Llama-3.1-8B	128,000	51	52	0.080%
Qwen2.5-7B	151,643	51	52	0.067%
Mistral-7B	32,768	36	40	0.226%
Gemma-2-9B	256,000	48	52	0.038%

Table 26: AUC comparison: logit-based metrics vs. BoW lexicon proxies (label = `is_success_llm`; stochastic samples excluded).

Method	All Models ($n = 2,040$)	Llama-only ($n = 840$)
S_0 (logit, real-time)	0.630	0.702
$\tilde{S}_{1.5}$ (logit, real-time)	0.701	0.735
$\tilde{S}$ (logit, real-time)	0.691	0.787
BoW — handcrafted lexicon	0.674	0.774
BoW — induced lexicon	0.996	0.936

out sensitivity, digit-token removal, whitespace-prefixed-token variants, and response-onset-token removal to test whether LMS is dominated by a small number of tokenizer-specific fragments.

F.9 Probe-Derived Lexicon Check

Probe-derived vocabulary rankings confirm that many handcrafted tokens align with model-internal refusal/compliance directions. We project vocabulary embeddings onto a refusal direction and report representative rank percentiles for single-token lexicon entries.

The ranking is not perfect, especially for compliance tokens, but the direction is consistent enough to support the handcrafted lexicon as an interpretable lower-access choice.

F.10 Vocabulary Top- k Logit Access

The diagnostic is hidden-state-free, but it is not fully black-box under APIs that expose only a small top- k set of output log-probabilities. A key practical question is whether S_t can be computed from the limited log-probability information returned by closed-source APIs. For example, an API may expose only a small `top_logprobs` window rather than the full vocabulary distribution. To test this directly, we simulate vocabulary-level top- k truncation on Llama-3.1-8B-Instruct across MCM, GCG, and DI (60 samples per attack).

Method. For each decoding step t , we take the full logit vector $\ell_t \in \mathbb{R}^{|\mathcal{V}|}$ and construct a truncated vector $\hat{\ell}_t^{(k)}$ by retaining only the k highest-valued entries and setting all others to $-\infty$. We then recompute S_t from $\hat{\ell}_t^{(k)}$, counting only lexicon tokens that survive in the top- k window. We evaluate $k \in \{5, 10, 20, 50, 100, 200\}$ and measure Pearson correlation with the full-logit S_t trajectory, mean absolute error (MAE), sign agreement, and refusal/compliance lexicon survival rates. This is distinct from Appendix F.6: lexicon-internal aggregation assumes the lexicon token logits are available, whereas vocabulary-level truncation asks whether those safety-relevant tokens appear in the API-visible top- k window at all.

Results. The results reveal a large gap between lexicon-internal aggregation stability and vocabulary-level truncation. At $k = 20$, a typical API-like level, only 2–3% of refusal lexicon tokens and 1.5–2.6% of compliance tokens survive in the top- k window. Trajectory correlation with the full-logit S_t trajectory is near zero or negative ($r = -0.120$ for MCM, $r = -0.009$ for GCG, and $r = -0.072$ for DI), and sign agreement is only 0.523–0.576. Even at $k = 200$, GCG remains near zero ($r = 0.042$), consistent with adversarial suffix optimization dispersing probability mass across unusual token positions that rarely appear in the top- k window.

Table 27: Top induced compliance words and AUC by attack paradigm (BoW-induced vs. BoW-handcrafted).

Attack	Top Induced Compliance Words	BoW-HC	BoW-Ind	ASR
MCM	<i>stealth, sedative, narrative, personnel</i>	0.730	0.977	10.2%
GCG	<i>blade, credentials, posts, anal</i>	0.537	0.969	46.7%
DI	<i>exploiting, injection, scam, deepfake</i>	0.500	0.971	62.8%

Table 28: Cross-attack transfer of induced lexicons ($n = 660$, all models). Diagonal entries (*) are in-distribution; off-diagonal entries are out-of-distribution.

Train → Test	DI	GCG	MCM
DI	0.891*	0.670	0.530
GCG	0.688	0.932*	0.676
MCM	0.606	0.734	0.890*

Implications. This simulation confirms that the current formulation requires full logits or targeted access to the relevant lexicon token log-probabilities. Standard top- k log-probability outputs are insufficient for reliable LMS computation. This empirically supports positioning S_t as a diagnostic for open-weight model auditing rather than a universal deployment-time detector, and motivates future work on API-compatible reformulations such as targeted logit-bias probing or lightweight distilled monitors.

F.11 Compliance-Dominant Successful Jailbreaks

Successful harmful generations are usually compliance-dominant in the aggregate signal, which is why LMS should be read temporally rather than as a one-shot polarity classifier. In the primary 12-condition grid, 92.3% of successful jailbreaks (263/285) have $\bar{S} > 0$, compared with 49.7% of failed jailbreaks (216/435). Table 32 reports the successful-jailbreak breakdown by condition.

G Judge and Behavioral Probe Checks

G.1 Judge Scope

The authoritative 12-condition ledger in Appendix A.1 uses Llama-Guard-3-8B as the primary judge. HarmBench-Mistral-7B and GPT-4o have also been refreshed on the same 12-condition grid. The auxiliary judges support the sample-level prediction results but show condition-level label disagreements, so the main evidence should be read as primary-judge evidence rather than a full judge-invariance claim.

HarmBench disagrees with the primary RP-plane annotations on Mistral+MCM and Qwen+MCM, both cases where refreshed judge labels yield axis-dominant PGB while the primary Llama-Guard permutation stage assigns NGB. GPT-4o agrees on 8 of 11 available annotations; its disagreements are Mistral+MCM and Qwen+MCM as PGB, plus Qwen+DI as IGB, while the primary judge assigns all three to NGB.

G.2 Adjudicated Two-Rater Human Audit

We additionally constructed a 240-item adjudicated human reference audit using two internal author-raters. The two raters independently labeled each response as safe/refusal, ambiguous/partial, or harmful compliance; all disagreements were then reconciled into a final adjudicated label. The audit intentionally oversamples automatic-judge disagreements and balanced success/failure cases, so its label proportions are not ASR estimates. Instead, the audit measures label reliability and compares binary judges against a human-adjudicated reference subset.

After adjudication, 132 items have non-ambiguous safe/refusal or harmful-compliance labels and can be directly compared with binary judges. On this adjudicated non-ambiguous subset, GPT-4o matched the human label on 103/132 items, Llama-Guard on 102/132 items, and HarmBench on

Table 29: Cross-model transfer of induced lexicons ($n = 660$, all attacks). Diagonal entries (*) are in-distribution.

Train → Test	Gemma	Llama	Mistral	Qwen
Gemma	0.970*	0.817	0.662	0.544
Llama	0.673	1.000*	0.705	0.682
Mistral	0.594	0.729	0.968*	0.692
Qwen	0.615	0.522	0.642	0.957*

Table 30: Probe-derived rank percentiles for representative handcrafted tokens. Higher values indicate stronger alignment with the intended refusal or compliance side.

Token	Llama	Mistral	Qwen	Gemma
<i>Refusal tokens, percentile from top</i>				
illegal	85.7%	100.0%	65.9%	99.4%
harmful	91.4%	98.1%	97.7%	99.2%
dangerous	64.9%	100.0%	77.4%	99.2%
Sorry	89.0%	89.4%	60.8%	50.3%
<i>Compliance tokens, percentile from bottom</i>				
Sure	90.0%	37.5%	51.1%	46.9%
Step	90.3%	43.6%	95.5%	78.6%
Below	50.6%	98.9%	93.0%	96.4%
Procedure	96.6%	49.3%	70.6%	33.5%

88/132 items. Thus the human audit supports the paper’s judge-conditioned framing: LLM-as-judge labels are useful response-level outcome variables, but borderline partial-compliance outputs remain a material source of label ambiguity rather than a solved reference-label problem.

G.3 Primary-Judge Intervention Summary

The early-stopping procedure is used as a behavioral probe, not as a proposed complete defense. The primary 12-condition Llama-Guard run is reported in Table 1: aggregate ASR decreases from 39.6% to 13.1% under the $w = 5$ probe. Because the behavioral probe is defined from t_{cross} , it is not independent of the temporal diagnostic.

G.4 S_0 -Gated Stress Test

The S_0 -gated variant is a stress test for attack-formatted benign references in the same 12-condition grid. Format-free benign queries have 0.0% format-free benign FPR in a separate check; attack-formatted benign references are harder because they intentionally share the attack wrapper. Thus, the table below should be read as wrapper-confound stress behavior rather than as the format-free benign FPR. Formally, the original halt rule is $H_w = \mathbf{1}[t_{\text{cross}} > w]$, and the gated rule is $H_{w,\tau}^{S_0} = \mathbf{1}[t_{\text{cross}} > w \wedge S_0 < \tau]$.

H Reproducibility Resources

H.1 Supplementary Archive

The supplementary archive provides benchmark input prompts, prompt-construction files, LMS extraction code, reproduction scripts, aggregate result files, and redacted per-sample metric tables for the main numerical claims. The prompt files support rerunning LMS extraction from model logits, while the redacted sample tables remove prompt text, generated responses, raw judge outputs, and local experiment folder names while preserving the metrics and labels needed to recompute the reported AUC, temporal, RP, hidden-alignment, and early-stopping summaries. The archive also includes a README with reproduction commands, lightweight and generation-specific Python requirements, and an external-asset license note.

Table 31: Vocabulary-level top- k truncation simulation (Llama-3.1-8B-Instruct, $n = 60$ per attack). Pearson r is the correlation between full-logit and truncated S_t trajectories. Sign Agr. is the fraction of steps preserving S_t polarity. Ref/Cmp Surv. is the mean fraction of refusal/compliance lexicon tokens surviving in the top- k window.

Attack	k	Pearson r	MAE	Sign Agr.	Ref Surv.	Cmp Surv.
MCM	5	-0.199	5.023	0.637	1.2%	0.6%
	10	-0.108	4.779	0.661	1.9%	1.0%
	20	-0.120	4.254	0.576	2.7%	1.9%
	50	0.111	2.899	0.576	5.1%	3.8%
	100	0.319	2.503	0.586	7.5%	5.4%
	200	0.472	2.236	0.619	10.6%	7.5%
GCG	5	-0.106	4.489	0.477	1.5%	0.8%
	10	-0.071	4.442	0.520	2.3%	1.5%
	20	-0.009	3.817	0.564	3.2%	2.6%
	50	-0.059	3.406	0.538	5.2%	4.3%
	100	-0.051	3.108	0.541	7.6%	6.3%
	200	0.042	2.674	0.588	10.9%	8.5%
DI	5	0.121	5.638	0.470	0.7%	0.5%
	10	-0.089	7.107	0.445	1.5%	0.8%
	20	-0.072	6.022	0.523	2.0%	1.5%
	50	0.117	4.651	0.612	2.9%	3.1%
	100	0.231	3.561	0.670	4.3%	4.6%
	200	0.387	2.773	0.673	6.5%	6.5%

Table 32: Successful jailbreaks with compliance-dominant generation ($\bar{S} > 0$) in the 12-condition grid.

Model	Attack	n_{succ}	$n_{\bar{S} > 0}$	Rate
Gemma	DI	23	23	100.0%
Gemma	GCG	20	20	100.0%
Gemma	MCM	11	11	100.0%
Llama	DI	33	30	90.9%
Llama	GCG	5	5	100.0%
Llama	MCM	3	3	100.0%
Mistral	DI	48	48	100.0%
Mistral	GCG	34	31	91.2%
Mistral	MCM	3	3	100.0%
Qwen	DI	35	35	100.0%
Qwen	GCG	46	42	91.3%
Qwen	MCM	24	12	50.0%
All		285	263	92.3%

H.2 Compute Resources

All reported main-grid generations and hidden-state captures were run on a single local workstation. The machine used one NVIDIA GeForce RTX 4090 GPU with 24 GB VRAM (CUDA 13.1, driver 590.48), an AMD Ryzen 9 3950X CPU with 16 cores and 32 threads, 62 GB system RAM, and a 457 GB root disk. Idle post-experiment readings were approximately 1.4 GB GPU memory in use and 60°C GPU temperature.

Gemma took longer primarily because Gemma-2-9B has more transformer layers (42) and therefore larger hidden-state outputs. Using the observed experiment-time GPU power draw of approximately 239 W, the total run consumed roughly $239\text{W} \times 3,480\text{s} \approx 832\text{ kJ}$, or 0.23 kWh. The main capture covered four models, 240 generations per model, about 64 decoding steps per generation, and three retained analysis layers, yielding approximately 184,320 recorded layer-step measurements in addition to the standard logit outputs.

Table 33: Primary and refreshed auxiliary judge summary on the same 12-condition grid. Success-label agreement is harmful-sample agreement with Llama-Guard. PGB/IGB/NGB annotation match counts condition-level agreement after the same permutation rule, excluding auxiliary conditions with insufficient successful samples.

Judge	ASR	Success-label agreement	$\bar{S}_{1:5}$ AUC	t_{cross} AUC	Annotation match
Llama-Guard	39.6%	–	0.771	0.784	12/12
HarmBench	33.3%	84.3%	0.752	0.734	10/12
GPT-4o	42.9%	87.8%	0.820	0.804	8/11 [†]

[†]Llama+MCM has only two GPT-4o successes and is reported as N/A for auxiliary RP-plane annotation.

Table 34: Adjudicated two-rater human audit of response-level jailbreak-success labels. Human labels use three categories: safe/refusal, ambiguous/partial, and harmful compliance. Binary harmful-compliance agreement collapses safe/refusal and ambiguous/partial into the non-compliance side.

Quantity	Value
Items labeled by both raters	240
Exact three-class agreement	193/240 (80.4%)
Cohen’s κ (three-class)	0.698
Binary harmful-compliance agreement	213/240 (88.8%)
Cohen’s κ (binary harmful-compliance)	0.732
Disagreements reconciled	47/47 (100.0%)
Final adjudicated labels	74 safe / 108 ambiguous / 58 harmful

H.3 External Assets and Licenses

Table 38 lists the external assets used in the experiments and the license or access terms reported by their official model cards or repositories. We use these assets for evaluation only and do not redistribute third-party model weights or service outputs in this submission. The supplementary archive includes benchmark input prompts for reproducibility under the license terms listed in Table 38.

Table 35: Early-stopping behavioral probe response by coarse RP-plane annotation. The response pattern shows behaviorally distinct responses, not an independent source of evidence; format-free benign FPR reports the separate natural-benign query check.

Annotation	Original ASR	Probe ASR	Δ ASR	Format-free benign FPR	Reading
PGB	33.3%	19.4%	13.9 pp	0.0%	Small/selective effect
IGB	28.8%	10.0%	18.8 pp	0.0%	Clear generation-time effect
NGB	52.0%	11.7%	40.3 pp	0.0%	Large but least diagnostic/selective

Table 36: S_0 -gated vs. original early stopping on attack-formatted benign reference stress tests in the 12-condition grid ($w = 5, \tau = 6.3$).

	Original		S_0 -Gated	
	Δ ASR	Attack-formatted false halt	Δ ASR	Attack-formatted false halt
Aggregate	26.5 pp	46.0%	17.1 pp	12.6%
Gemma	20.0 pp	37.8%	10.6 pp	0.0%
Llama	17.8 pp	52.8%	14.4 pp	6.1%
Mistral	27.8 pp	45.6%	27.8 pp	44.4%
Qwen	40.6 pp	47.8%	16.1 pp	0.0%

Table 37: Approximate runtime for the main per-model evaluation runs, estimated from output file timestamps.

Run	Completion time	Approximate runtime
Llama-3.1-8B-Instruct	22:50	12 min
Qwen2.5-7B-Instruct	23:02	12 min
Mistral-7B-Instruct-v0.3	23:14	12 min
Gemma-2-9B-Instruct	23:36	22 min
Analysis scripts	23:36	< 1 min
Total	–	58 min

Table 38: External assets used in the experiments.

Asset	Use in this paper	License or terms
Meta-Llama-3.1-8B-Instruct	Target model	Llama 3.1 Community License and acceptable-use terms
Llama-Guard-3-8B	Primary LLM-as-judge safety classifier	Llama 3.1 Community License and acceptable-use terms
Mistral-7B-Instruct-v0.3	Target model	Apache License 2.0
Qwen2.5-7B-Instruct	Target model	Apache License 2.0
Gemma-2-9B-Instruct	Target model	Gemma license / Gemma Terms of Use
JailbreakBench JBB-Behaviors	Harmful prompt source	MIT License
HarmBench-Mistral-7B validation classifier	Auxiliary LLM-as-judge safety classifier	MIT License
GPT-4o	Auxiliary judge refresh	OpenAI service access terms; no weights redistributed